

Polyhedral Aspects of the Stable Set Problem in Relation to Odd Cycles

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I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references.

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Abstract

We establish properties of the stable set polytope for graphs that have few odd cycles in two different senses: First, we consider graphs for which the number of vertex-disjoint odd cycles is bounded by a constant. Second, we consider graphs which have an odd cycle transversal, i. e. a subset of vertices that intersects any odd cycle, whose size can be bounded by a constant.

We show that for any fixed positive ε , the multiplicative integrality gap of all standard 0/1-integer programming hierarchies is at most $1 + \varepsilon$ after a constant number of rounds for the first type of graphs, and for the second type is 1 after a constant number of rounds. In particular, this allows us to determine the optimal value, up to a factor of $1 + \varepsilon$, in polynomial time for the weighted stable set problem on graphs of the first type and yields a polynomial algorithm for the second type, with both of the algorithms being purely polyhedral.

As a key ingredient, we show a graph theoretic result that is of independent interest: Any graph with vertex weights and no odd cycle of weight less than ε times the total weight of all vertices has an odd cycle transversal of size at most $c_1/\varepsilon \ln(c_2/\varepsilon)$ for explicit constants c_1, c_2 , improving a result of Bock et al. [1].

Moreover, we provide an overview of some classes of graphs for which a complete linear description of their stable set polytopes is known, as well as of families of valid constraints for the stable set polytope. By specializing a result of Shepherd [2], we obtain a complete linear description of the stable set polytope of graphs that have an odd cycle transversal of size at most three and that are the complement of so-called quasi-line graphs. Furthermore, we show that the CG rank of the fractional stable set polytope can be bounded by the size of an odd cycle transversal in this case.

Lastly, we examine the role of the constraints that define the stable set polytope of complements of quasi-line graphs with an odd cycle transversal number of 2 for the stable set polytopes of general graphs with an odd cycle transversal of size 2, using computational experiments with random graphs. The results show that the constraint classes relevant in the restricted case explain many, but not all, facets in the general case, with the share of unexplained facets increasing as graph sizes increase.

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1 Introduction

For a graph $G = (V, E)$, a *stable set* (also called *independent set* or *coclique*) is a subset of vertices that are pairwise non-adjacent. The *stable set problem* is to determine a stable set of maximum cardinality, or, for given weights on the vertices, a stable set of maximum total weight. The stable set problem is well known to be NP hard, even in the unweighted version [3].

One approach to the stable set problem is to define a polytope that captures the structure of all stable sets in a graph and then use polyhedral methods to reason about it. To this end, consider the polytope given by the convex hull of all characteristic vectors of stable sets. This polytope is called the *stable set polytope* and we formally define it as

$$\text{STAB}(G) = \text{conv}\{\chi(S) \in \{0, 1\}^V : S \subseteq V, S \text{ is a stable set}\},$$

where $\chi(S)$ denotes the characteristic vector of S , i. e. $(\chi(S))_v = 1$ if $v \in S$ and 0 otherwise. Note that solving the stable set problem can equivalently be phrased as solving the optimization problem

$$\max \sum_{v \in V} c_v x_v \quad \text{s. t. } x \in \text{STAB}(G), \quad (1.1)$$

where $c_v = 1$ for all $v \in V$ in the unweighted case and c_v is the weight of vertex v in the weighted case for all $v \in V$. However, to make this problem tractable, we would need a concise description of $\text{STAB}(G)$ by a set of linear inequalities. Due to the hardness of the problem, such a description most likely does not exist. Nevertheless, this approach can be utilized to achieve interesting results. The first step is to observe that

$$\text{STAB}(G) = \text{conv}\{x \in \{0, 1\}^V : x_v + x_w \leq 1 \text{ for all } \{v, w\} \in E\}.$$

Replacing the binary constraints by $x \in [0, 1]^V$ gives rise to another polytope which we call the *fractional stable set polytope* and denote it by

$$\text{FSTAB}(G) = \{x \in [0, 1]^V : x_v + x_w \leq 1 \text{ for all } \{v, w\} \in E\}.$$

The constraints used here are called *edge-constraints* and we will revisit them later.

Note that $\text{STAB}(G) = \text{conv}(\text{FSTAB}(G) \cap \mathbb{Z}^V)$. For a general polyhedron $P \subseteq \mathbb{R}^n$, we call $P_I := \text{conv}(P \cap \mathbb{Z}^n)$ the *integer hull* of P and we call any polyhedron $Q \in \mathbb{R}^n$ with $Q_I = P_I$ a *relaxation* of P_I . Consequently, we can rephrase the relationship of $\text{STAB}(G)$ and $\text{FSTAB}(G)$ by saying that $\text{FSTAB}(G)$ is a relaxation of $\text{STAB}(G)$. Given this relaxation, we can consider the relaxed optimization problem

$$\max \sum_{v \in V} c_v x_v \quad \text{s. t. } x \in \text{FSTAB}(G). \quad (1.2)$$

Note that this optimization problem can be solved efficiently using polynomial time algorithms for linear programming.

Of course, the optimal solutions and the optimal value of (1.1) and (1.2) are not the same in general, and they can differ by a lot. Actually, it is well known that $\text{FSTAB}(G) = \text{STAB}(G)$ if and only if G is bipartite, or equivalently if G has no odd cycles. In particular, this means that for graphs without odd cycles, the weighted stable set problem is solvable in polynomial time. As a generalization of this fact, Fiorini, Joret, Weltge, and Yuditski [4] were recently able to show that the weighted stable set problem is polynomial time solvable for any graph G that has at most k vertex-disjoint odd cycles, where k is a constant. This motivates the definition of the *odd cycle packing number* $\text{ocp}(G)$ as the largest number of vertex-disjoint odd cycles in G , which somehow seems to be related to the difficulty of the stable set problem.

A related concept to the odd cycle packing number is the minimal size of a subset T of vertices such that any odd cycle contains a vertex of T . We call such a subset an *odd cycle transversal* and call the size of a minimal odd cycle transversal in a graph G the *odd cycle transversal number* $\text{oct}(G)$.

Here, we want to mention that both graphs with bounded odd cycle packing number and graphs with bounded odd cycle transversal number could justifiably be called *nearly bipartite*, and this term has been used for graphs with bounded odd cycle transversal number before [5]. However, to avoid privileging one notion over the other and to avoid confusion with the notion of being *k-near bipartite* introduced in [6] and the distinct notion of *near-bipartite* used in [2], we will avoid using this or similar terms.

We also want to highlight that there is a different, non-combinatorial approach to defining the odd cycle packing number. Using the edge-node-incidence matrix $I_G \in \{0, 1\}^{E \times V}$, i. e. $(I_G)_{e,u} = 1$ if u is incident to e and 0 otherwise, we can rewrite $\text{FSTAB}(G)$ compactly as

$$\text{FSTAB}(G) = \{x \in [0, 1]^V : I_G x \leq \mathbf{1}\}$$

where $\mathbf{1} = (1, \dots, 1)^T \in \{0, 1\}^E$ denotes the all-ones vector. It turns out that the largest subdeterminant of I_G is $2^{\text{ocp}(G)}$ [7]. Thus, instead of considering graphs with bounded odd cycle packing number, we can equivalently consider graphs for which the coefficient matrix describing the fractional stable set polytope has bounded subdeterminants. In fact, this is the approach originally taken by Fiorini et al. [4]. However, we will not delve deeper into the connection to subdeterminants in this thesis.

In the following chapter, we will provide an overview of so-called *hierarchies of relaxations* of 0/1-polytopes and establish first results that connect these to odd cycle transversals. Afterwards, in Chapter 3, we will show that graphs without short odd cycles have small odd cycle transversals, for two notions of short odd cycles. In the subsequent chapter, we will then use this result to lift a result of the second chapter to the case of graphs with bounded odd cycle packing number. In Chapter 5, we will present an overview of some classes of graphs for which a complete linear description of their stable set polytopes is known and an overview of families of valid constraints for the stable set polytope. Lastly, in the final chapter, we computationally examine the stable set polytope of graphs with odd cycle transversal number two.

2 Hierarchies of relaxations for 0/1-polytopes

As mentioned before, one way to approach the integer hull P_I of a polytope $P \subseteq [0, 1]^n$ is through relaxations. Multiple hierarchies of relaxations have been proposed that construct a nested sequence of polytopes (or sometimes also more general convex bodies) of iterates $P^{(i)}$ that satisfy

$$P = P^{(0)} \subseteq P^{(1)} \subseteq \dots \subseteq P^{(r)} = P_I$$

for some $r \in \mathbb{Z}_{\geq 0}$ and therefore provide intermediate steps between P and P_I .

2.1 Balas-Ceria-Cornuéjols hierarchy

To introduce the first such hierarchy, we define the *partial convexification* as follows.

Definition 2.1. For a polytope $P \subseteq [0, 1]^n$ and $S \subseteq \{1, \dots, n\} := [n]$ let

$$P^{(S)} = \text{conv}(P \cap \{x \in \mathbb{R}^n : x_i \in \{0, 1\} \text{ for } i \in S\}).$$

We call $P^{(S)}$ the partial convexification of P with regard to S .

Regarding the naming, observe that the integer hull $P_I = \text{conv}(P \cap \mathbb{Z}^n)$ is the convexification of the set of integer points $P \cap \mathbb{Z}^n$ and P_I can also be viewed as $P^{([n])}$. Hence, the objects obtained for a subset of $[n]$ are called a partial convexification.

Note that for a chain $\emptyset = I_0 \subseteq I_1 \subseteq \dots \subseteq I_r = [n]$ and by setting $P^{(j)} = P^{(I_j)}$, we have

$$P = P^{(0)} \subseteq P^{(1)} \subseteq \dots \subseteq P^{(r)} = P_I,$$

meaning that this construction indeed yields a hierarchy of relaxations.

Balas, Ceria, and Cornuéjols [8] first considered hierarchies of this type for the special case where in each step exactly one variable is added. They are therefore sometimes called *Balas-Ceria-Cornuéjols hierarchies* [9]. Moreover, they observed that these hierarchies can actually be computed in a simpler way. To this end, they proved the following theorem.

Theorem 2.2. For $S = \{i_1, \dots, i_k\} \subseteq [n]$ we have $P^{(S)} = \left(\left((P^{(\{i_1\})})^{(\{i_2\})} \right) \dots \right)^{(\{i_k\})}$.

Now, to obtain an iterate of a Balas-Ceria-Cornuéjols hierarchy, one can follow an approach of *sequential convexification*: Instead of taking the partial convexification with regard to all variables in I_j to obtain $P^{(j)}$, one can sequentially take the partial convexification of $P^{(j-1)}$ with regard to the one additional variable in $I_j \setminus I_{j-1}$.

We present the proof given by Conforti et al. [10]. For it, we need the following lemma.

Lemma 2.3. Let $T \subseteq \mathbb{R}^n$ and $a \in \mathbb{R}^n \setminus \{0\}$, $\beta \in \mathbb{R}$. Consider the hyperplane $H = \{x : a^\top x = \beta\}$ and the halfspace $H^\leq = \{x : a^\top x \leq \beta\}$. If $T \subseteq H^\leq$, then $\text{conv}(T \cap H) = \text{conv}(T) \cap H$.

Proof. Since $T \cap H \subseteq \text{conv}(T) \cap H$ and the right-hand side is convex, we have $\text{conv}(T \cap H) \subseteq \text{conv}(T) \cap H$. Conversely, if $x \in \text{conv}(T) \cap H$ then $x = \sum_{i=1}^m \lambda_i x_i$ for some $m \in \mathbb{Z}_{\geq 1}$, $\lambda_i \geq 0$, $x_i \in T$ for $i \in [m]$ and $\sum_{i=1}^m \lambda_i = 1$. Since $x \in H$ and $x_i \in T \subseteq H^\leq$, $x_i \in H$ for $i \in [m]$ and therefore $x \in \text{conv}(T \cap H)$. $\square$

Proof of Theorem 2.2. Without loss of generality $S = [k]$. We prove the result by induction on k . Let $Q_S = P \cap \{x \in \mathbb{R}^n : x_i \in \{0, 1\} \text{ for } i \in S\}$, so that $P^{(S)} = \text{conv}(Q_S)$ by definition.

The case $k = 1$ is trivial. For the inductive step, first observe that

$$\begin{aligned} P^{([k-1])} \cap \{x \in \mathbb{R}^n : x_k = i\} &= \text{conv}(Q_{[k-1]}) \cap \{x \in \mathbb{R}^n : x_k = i\} \\ &= \text{conv}(Q_{[k-1]} \cap \{x \in \mathbb{R}^n : x_k = i\}) \end{aligned}$$

for $i \in \{0, 1\}$ by the definition of $Q_{[k-1]}$ and by Lemma 2.3. We can use this to show that

$$\begin{aligned} \left(P^{([k-1])}\right)^{(\{k\})} &= \text{conv}\left(\left(P^{([k-1])} \cap \{x \in \mathbb{R}^n : x_k = 0\}\right) \cup \left(P^{([k-1])} \cap \{x \in \mathbb{R}^n : x_k = 1\}\right)\right) \\ &= \text{conv}\left(\text{conv}(Q_{[k-1]} \cap \{x \in \mathbb{R}^n : x_k = 0\}) \cup \text{conv}(Q_{[k-1]} \cap \{x \in \mathbb{R}^n : x_k = 1\})\right) \\ &= \text{conv}\left((Q_{[k-1]} \cap \{x \in \mathbb{R}^n : x_k = 0\}) \cup (Q_{[k-1]} \cap \{x \in \mathbb{R}^n : x_k = 1\})\right) \\ &= P^{([k])} \end{aligned}$$

where the first and last equality hold by definition and the third equality holds since

$$\text{conv}(\text{conv}(A) \cup \text{conv}(B)) = \text{conv}(A \cup B)$$

for any $A, B \subseteq \mathbb{R}^n$. Finally, by the induction hypothesis

$$\left(\left(\left(P^{(\{1\})}\right)^{(\{2\})}\right) \dots\right)^{(\{k\})} = \left(P^{([k-1])}\right)^{(\{k\})}.$$

□

Moreover, Balas et al. [8] showed that $P^{(S)}$ can be written as the projection of a polytope in a higher-dimensional space that has an inequality description at most $2^{|S|}$ times the size of an inequality description of P . Writing a polytope as such a projection of a higher-dimensional polytope is known as an *extended formulation*, see, e.g., [11] for an introduction to extended formulations. In particular, this implies the following theorem.

Theorem 2.4. *For a polytope $P \subseteq [0, 1]^n$ given by an inequality description with encoding length L and any $S \subseteq [n]$ of fixed size, it is possible to maximize any linear objective over $P^{(S)}$ in polynomial time where polynomial refers to L and the encoding length of the objective.*

Before we discuss the application of this hierarchy to the fractional stable set polytope, we first need to introduce some useful properties that it has. To this end, we define *monotone polyhedra* as follows.

Definition 2.5. *A polyhedron $P \subseteq [0, 1]^n$ is called monotone, if for any $x, y \in \mathbb{R}^n$ with $x \in P$ and $0 \leq y \leq x$ it also holds that $y \in P$.*

We can characterize monotone polyhedra in a neat way as follows.

Lemma 2.6. *A full-dimensional polyhedron $P \subseteq [0, 1]^n$ is monotone, if and only if there is some $m \in \mathbb{Z}_{\geq 0}$ and $A \in \mathbb{R}_{\geq 0}^{m \times n}, b \in \mathbb{R}^m$ such that $P = \{x \in [0, 1]^n : Ax \leq b\}$.*

Proof. Assume there is $m \in \mathbb{Z}_{\geq 0}, A \in \mathbb{R}_{\geq 0}^{m \times n}, b \in \mathbb{R}^m$ such that $P = \{x \in [0, 1]^n : Ax \leq b\}$. Let $x, y \in \mathbb{R}^n$ with $x \in P$ and $0 \leq y \leq x$. Clearly, we have $y \in [0, 1]^n$. Moreover, since A has non-negative entries, we have $Ay \leq Ax \leq b$. Therefore, $y \in P$.

For the converse direction, let $a \in \mathbb{R}^n, \beta \in \mathbb{R}$ such that $a^\top x \leq \beta$ is a facet-inducing inequality of P and let $F = P \cap \{x : a^\top x = \beta\}$ be the corresponding facet. We consider two cases. First, if $F \subseteq \{x : x_i = 0\}$ for some $i \in [n]$, we must have $F = \{x : x_i = 0\} \cap P$, since P is full-dimensional, and therefore $a^\top x \leq \beta$ is a scalar multiple of $x_i \geq 0$. Second, if $F \not\subseteq \{x : x_i = 0\}$ for all $i \in [n]$, then for all $i \in [n]$ there is $x^{(i)} \in F$ with $x_i^{(i)} > 0$. Let $(y^{(i)})_j = x_j^{(i)}$ for $j \in [n] \setminus \{i\}$ and let $(y^{(i)})_i = 0$. Since P is monotone $y^{(i)} \in P$ and therefore $a^\top y^{(i)} \leq \beta$. It follows that $0 \leq \beta - a^\top y^{(i)} = a^\top x^{(i)} - a^\top y^{(i)} = a_i x_i^{(i)}$ and hence $a_i \geq 0$. This means that collecting all facet-inducing inequalities yields non-negativity constraints and a linear system with a non-negative constraint matrix, which is the claimed description. □

The partial convexification is structure-preserving for monotone polyhedra in the following way.

Lemma 2.7. *For any monotone polyhedron $P \subseteq [0, 1]^n$ and any set $S \subseteq [n]$, the partial convexification $P^{(S)}$ with regard to S is also monotone.*

Proof. By Theorem 2.2, it is sufficient to consider the case $S = \{i\}$. Let $x \in P^{(\{i\})}$, then there are $x^{(l)} \in P \cap \{x : x_i = l\}$ for $l \in \{0, 1\}$ such that $x \in \text{conv}\{x^{(0)}, x^{(1)}\}$.

Consider any $j \in [n]$ and any $\mu \in [0, x_j]$. We will show that $x - \mu e_j \in P^{(\{i\})}$. To this end, for any $y \in \mathbb{R}_{\geq 0}^n$ let $y_{\downarrow}$ denote the vector with $(y_{\downarrow})_k = y_k$ for $k \neq j$ and $(y_{\downarrow})_j = 0$. Then $x_{\downarrow} \in \text{conv}\{x_{\downarrow}^{(0)}, x_{\downarrow}^{(1)}\}$ and $x - \mu e_j \in \text{conv}\{x, x_{\downarrow}\}$. Since P is monotone and $0 \leq x_{\downarrow}^{(l)} \leq x^{(l)}$, $x_{\downarrow}^{(l)} \in P$ for $l \in \{0, 1\}$. Moreover, since $(x_{\downarrow}^{(l)})_i = x_i^{(l)} = l$ if $i \neq j$ and $(x_{\downarrow}^{(l)})_i = 0$ if $i = j$, we even have $x_{\downarrow}^{(l)} \in P^{(\{i\})}$ for $l \in \{0, 1\}$. We conclude $x - \mu e_j \in \text{conv}\{x, x_{\downarrow}\} \subseteq \text{conv}\{x^{(0)}, x^{(1)}, x_{\downarrow}^{(0)}, x_{\downarrow}^{(1)}\} \subseteq P^{(\{i\})}$.

Now let $y \in \mathbb{R}^n$ with $0 \leq y \leq x$. By sequentially reducing each component and using the consideration above, we conclude $y \in P^{(\{i\})}$. $\square$

Equipped with these results, we can proceed to apply the partial convexification to the fractional stable set polytope and obtain a first simple result.

Theorem 2.8. *For any graph $G = (V, E)$ and any odd cycle transversal T in G , we have $\text{FSTAB}(G)^{(T)} = \text{STAB}(G)$.*

Proof. By definition, we have

$$\text{FSTAB}(G)^{(T)} = \text{conv} \left(\bigcup_{z \in \{0,1\}^T} (\text{FSTAB}(G) \cap \{x : x_T = z\}) \right),$$

where $x_T = z$ denotes $x_i = z_i$ for $i \in T$.

For any $z \in \{0, 1\}^T$, let $V_z = \bigcup_{v \in T: z_v = 1} N(v)$ where $N(v)$ denotes the neighborhood of v , i.e. all vertices adjacent to v in G , and let $G_z = G[V \setminus (V_z \cup T)]$. Note that $x \in [0, 1]^V$ with $x_T = z$ is in $\text{FSTAB}(G)$ if and only if $x_{V \setminus (V_z \cup T)}$ is in $\text{FSTAB}(G_z)$ and $x_{V_z} = \mathbf{0}$. Therefore

$$\begin{aligned} \text{FSTAB}(G) \cap \{x : x_T = z\} &= \{x \in \mathbb{R}^V : x_{V \setminus (V_z \cup T)} \in \text{FSTAB}(G_z), x_T = z, x_{V_z} = \mathbf{0}\} \\ &= \{x \in \mathbb{R}^V : x_{V \setminus (V_z \cup T)} \in \text{STAB}(G_z), x_T = z, x_{V_z} = \mathbf{0}\} \\ &\subseteq \text{STAB}(G), \end{aligned}$$

where the second equality follows, since G_z is a bipartite graph. From this, it follows that $\text{FSTAB}(G)^{(T)} \subseteq \text{STAB}(G)$ and the other inclusion is trivial. $\square$

By combining this result with Theorem 2.4, we see that for any constant k and any graph G with $\text{oct}(G) \leq k$, we can optimize over $\text{STAB}(G)$ in polynomial time. In particular, this implies that the stable set problem can be solved in polynomial time for graphs with odd cycle transversal number at most k . Note, that this result in itself is easy to prove by constructing a combinatorial algorithm that runs in polynomial time for such graphs. However, the algorithm that results from Theorem 2.8 and 2.4 is purely polyhedral and it is less obvious that there is a polyhedral algorithm to solve the stable set problem in polynomial time in the case of bounded odd cycle transversal number.

Next, we consider another way to look at the result. To this end, we introduce the following definition.

Definition 2.9. *For a polytope $P \in [0, 1]^n$ and a vector $c \in \mathbb{R}^n$, let*

$$\text{OPT}(P, c) = \max\{c^\top x : x \in P\}.$$

Moreover, for $S \subseteq [n]$, let

$$c(S) = \sum_{i \in S} c_i.$$

We can equivalently phrase Theorem 2.8 by stating that for any vector $c \in \mathbb{R}^V$ with at least one positive entry, the multiplicative integrality gap

$$\frac{\text{OPT}(\text{FSTAB}(G)^{(S)}, c)}{\text{OPT}(\text{STAB}(G), c)} = 1$$

for any graph G and any odd cycle transversal S in G . We will later show that for a special class of graphs, a partial convexification with regard to an odd cycle transversal of a carefully chosen subgraph has an integrality gap arbitrarily close to one. To show this result, we will need the following extension of Theorem 2.8 to the case where we only have an odd cycle transversal of an induced subgraph of G .

Lemma 2.10. *Let $G = (V, E)$ be a graph, $V' \subseteq V$ a subset of vertices, T an odd cycle transversal in $G' = G[V \setminus V']$, and let $c \in \mathbb{R}_{\geq 0}^V$ be a non-zero vector of vertex weights. Then, we have*

$$\frac{\text{OPT}(\text{FSTAB}(G)^{(T)}, c)}{\text{OPT}(\text{STAB}(G), c)} \leq 1 + \frac{c(V')}{\text{OPT}(\text{STAB}(G), c)}.$$

Proof. Let $c' = c_{V \setminus V'}$. If $c' = \mathbf{0}$, the claim is equivalent to

$$\text{OPT}(\text{FSTAB}(G)^{(T)}, c) \leq \text{OPT}(\text{STAB}(G), c) + c(V'),$$

which is clearly true. We can therefore assume $c' \neq \mathbf{0}$.

By repeated application of Lemma 2.3, we see that

$$\text{FSTAB}(G)^{(T)} \cap \{x : x_v = 0 \text{ for } v \in V'\} = \{x : x_{V \setminus V'} \in \text{FSTAB}(G')^{(T)}, x_{V'} = \mathbf{0}\}.$$

Let x^* be a solution attaining the optimal value $\text{OPT}(\text{FSTAB}(G)^{(T)}, c)$. By Lemma 2.6, $\text{FSTAB}(G)$ is a monotone polyhedron and by Lemma 2.7, so is $\text{FSTAB}(G)^{(T)}$. This means $\tilde{x}^*$ given by $(\tilde{x}^*)_v = 0$ for $v \in V'$ and $(\tilde{x}^*)_v = x_v^*$ for $v \notin V'$ satisfies $\tilde{x}^* \in \text{FSTAB}(G)^{(T)}$. Since it is also in $\{x : x_v = 0 \text{ for } v \in V'\}$, by the reasoning above, $\tilde{x}^* \in \{x : x_{V \setminus V'} \in \text{FSTAB}(G')^{(T)}, x_{V'} = \mathbf{0}\}$. We conclude

$$\text{OPT}(\text{FSTAB}(G)^{(T)}, c) = c^\top x^* \leq c^\top \tilde{x}^* + c(V') \leq \text{OPT}(\text{FSTAB}(G')^{(T)}, c') + c(V').$$

And therefore

$$\begin{aligned} \frac{\text{OPT}(\text{FSTAB}(G)^{(T)}, c)}{\text{OPT}(\text{STAB}(G), c)} &\leq \frac{\text{OPT}(\text{FSTAB}(G')^{(T)}, c') + c(V')}{\text{OPT}(\text{STAB}(G), c)} \\ &\leq \frac{\text{OPT}(\text{FSTAB}(G')^{(T)}, c')}{\text{OPT}(\text{STAB}(G'), c')} + \frac{c(V')}{\text{OPT}(\text{STAB}(G), c)} \\ &= 1 + \frac{c(V')}{\text{OPT}(\text{STAB}(G), c)} \end{aligned}$$

where we used $\text{OPT}(\text{STAB}(G'), c') \leq \text{OPT}(\text{STAB}(G), c)$ in the second step, and the last step follows from Theorem 2.8. $\square$

2.2 Chvátal-Gomory closure

A second hierarchy is based on so-called *cuts*, constraints that are valid for P_I but (potentially) cut off some points of P . Given an inequality $a^\top x \leq \beta$ valid for P with $a \in \mathbb{Z}^n$ and $\beta \notin \mathbb{Z}$ the stronger inequality $a^\top x \leq \lfloor \beta \rfloor$ is an example of such a cut. Inequalities obtained in this way are called *Chvátal-Gomory cuts* or *CG cuts* after Chvátal [12] and Gomory [13, 14] who introduced them.

We can construct a hierarchy from these cuts by applying all possible such cuts simultaneously and taking the result as the next iterate of the hierarchy. Formally, for a polyhedron $P \in \mathbb{R}^n$, we define

$$P' = \bigcap_{a \in \mathbb{Z}^n, \beta \in \mathbb{R}: a^\top x \leq \beta \ \forall x \in P} \{x \in \mathbb{R}^n : a^\top x \leq \lfloor \beta \rfloor\}$$

and we set $P^{(0)} = P$ and $P^{(i)} = (P^{(i-1)})'$ for $i \geq 1$. P' is called the *Chvátal-Gomory closure* of P or the *CG closure* for short. While the CG closure is an interesting object for theoretical considerations and can offer insight into the effectiveness of some cuts, it is a difficult object to handle algorithmically. Eisenbrand [15] showed that solving the separation problem for the CG closure is NP-hard, which, by the equivalence of optimization and separation [16], implies that optimizing over it is NP-hard.

Chvátal [17] showed that for any bounded polyhedron P there is some $r \in \mathbb{Z}_{\geq 0}$ such that $P^{(r)} = P_I$. Schrijver [18] extended the result to rational polyhedra, i. e. polyhedra that can be described by a system of inequalities with rational coefficients. The smallest $r \in \mathbb{Z}_{\geq 0}$ such that $P^{(r)} = P_I$ is called the *Chvátal-Gomory rank* or *CG-rank* of P . For an inequality $a^\top x \leq \beta$ that is valid for P_I the smallest $r \in \mathbb{Z}_{\geq 0}$ such that the inequality is valid for $P^{(r)}$ is called the *Chvátal-Gomory rank*, or the *CG rank*, of the inequality.

For general polyhedra, even in two dimensions, the CG-rank can be arbitrarily large. For polyhedra that are contained in the unit cube $[0, 1]^n$, however, Eisenbrand and Schulz [19] showed that the CG-rank is at most $\mathcal{O}(n^2 \log n)$. On the converse, Rothvoß and Sanità showed that it can be as large as $\Omega(n^2)$.

Note that if P is given by an inequality description $Ax \leq b$ for some $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$, any valid inequality (with the smallest possible right hand side) can be expressed as $\lambda^\top Ax \leq \lambda^\top b$ for some $\lambda \in \mathbb{R}_{\geq 0}^m$. Using this, we can give an equivalent description of the CG-closure

$$P' = \bigcap_{\lambda \in \mathbb{R}_{\geq 0}^m : \lambda^\top A \in \mathbb{Z}^n} \{x \in \mathbb{R}^n : \lambda^\top Ax \leq \lfloor \lambda^\top b \rfloor\}.$$

Of course, this hierarchy can be applied to the fractional stable set polytope. Pleasantly, it turns out that the CG-closure of the fractional stable set polytope has a simple explicit description, which we will return to in Section 5.2. Moreover, it is known that for general graphs, the CG rank of the fractional stable set polytope can be arbitrarily large. Chvátal et al. [20] give a family of graphs whose fractional stable set polytopes have CG rank $\Omega(n)$ where n is the number of nodes in the graph. However, for some graph classes, which we will see examples of later on, the CG rank is bounded by a constant.

2.3 Other hierarchies of relaxations

Several other hierarchies have been developed. For two hierarchies with iterates $P^{(0)}, \dots, P^{(r)}$, respectively $P^{(0)'}, \dots, P^{(r)'}$, we say that the first is *stronger* than the second, if $P^{(i)} \subseteq P^{(i)'}$ for all $i \in [r]$.

A few hierarchies that have been developed include the Lovász-Schrijver hierarchy (of which there is a linear and a semidefinite version), the Sherali-Adams hierarchy, and the Lasserre hierarchy. They are mentioned here in order of increasing strength and are all stronger than any Balas-Ceria-Cornuéjols hierarchy [9]. In particular, for all of them and any polytope $P \subseteq [0, 1]^n$, the n -th iterate coincides with the integer hull P_I , and any result that we prove for any iterate of a Balas-Ceria-Cornuéjols hierarchy also holds for the respective iterate of any of the previously mentioned hierarchies.

Moreover, all of these hierarchies share the property of Theorem 2.4 with the Balas-Ceria-Cornuéjols hierarchies: One can optimize a linear objective function over an iterate $P^{(i)}$ in polynomial time (under some assumptions on P) [9].

Lastly, all of these hierarchies are incomparable to the one derived from the CG closure, in the sense that neither is stronger than the other in general. The fact that the CG hierarchy cannot be stronger in general follows, for example, from the work of Rothvoß and Sanità [21], and the converse follows, for example, from the work of Kurpisz et al. [22].

3 Odd cycle transversals in graphs without short odd cycles

In Section 2.1 on Balas-Ceria-Cornuéjols hierarchies, we have seen results concerning properties of the fractional stable set polytope in terms of the odd cycle transversal number. This section establishes two results that guarantee the existence of a small odd cycle transversal under the condition that no short odd cycles are present. First, we prove such a result for unweighted graphs, where the length of a cycle is measured by the number of its vertices. Afterwards, we generalize the result to graphs with weighted vertices where the length of a cycle is measured by the total weight of its vertices.

In the remainder of the section, $G = (V, E)$ denotes a graph and $n = |V|$ denotes the number of vertices of G .

3.1 The unweighted case

Theorem 3.1. *Let $0 < \varepsilon \leq 1$ and let G have no odd cycles shorter than εn . Then $\text{oct}(G) \leq \frac{4}{\varepsilon} \ln\left(\frac{15}{\varepsilon}\right)$.*

Variants of this result were given by Komlós [23] and Györi et al. [5] as well as Har-Peled and Rahul [24]. Moreover, Komlós and Györi et al. showed that the given bound is tight up to constant factors.

In addition, using the results of the previous section, we see that for graphs of this class, the partial convexification of $\text{FSTAB}(G)$ with regard to a set of variables whose size only depends on ε is already equal to $\text{STAB}(G)$.

Corollary 3.2. *Let $0 < \varepsilon \leq 1$ and let G have no odd cycles shorter than εn . Then there is a set T of size $|T| \leq \frac{4}{\varepsilon} \ln\left(\frac{15}{\varepsilon}\right)$ for which $\text{FSTAB}(G)^{(T)} = \text{STAB}(G)$.*

Proof. This is an immediate consequence of Theorem 2.8 and Theorem 3.1. □

To prove Theorem 3.1, we will use the construction given by Komlós [23] and Györi et al. [5], analyze it following an adapted version of Komlós's [23] proof, and incorporate an idea from Har-Peled and Rahul [24] to improve constants. In doing so, to the best of our knowledge, we are able to improve upon the best previously known explicit constants.

The proof relies on the following Lemma.

Lemma 3.3. *Let $0 < \varepsilon \leq 1$, and let G be a non-bipartite graph which has no odd cycles shorter than $\varepsilon n \geq 10$. Then there is a nonempty subset of vertices $V' \subseteq V$ and $\alpha > 0$ such that $|V'| \leq \alpha \frac{4}{\varepsilon} \ln\left(\frac{15}{\varepsilon}\right)$ and removing V' from G yields a graph G' in which at most $(1 - \alpha)n$ vertices do not belong to bipartite components.*

Using this lemma, we can prove Theorem 3.1 by a simple induction.

Proof of Theorem 3.1. If $\varepsilon n < 10$, we have $\frac{4}{\varepsilon} \ln\left(\frac{15}{\varepsilon}\right) \geq \frac{4n}{10} \ln(15) \geq n$ and the claim is trivial.

We now prove the case $\varepsilon n \geq 10$ by induction on n . For $n \leq 10$ the claim is trivial since $\frac{4}{\varepsilon} \ln\left(\frac{15}{\varepsilon}\right) \geq 4 \ln(15) \geq 10 \geq n$. Assume now that $n > 10$. If G is already bipartite, there is nothing to show. Otherwise, let α be the number and $V' \subseteq V$ be the subset given by Lemma 3.3. Let G' be the graph that results from G by removing V' . Let G'' be the graph that results from G' by removing all vertices that belong to bipartite connected components. Since V' is nonempty, G'' has strictly fewer vertices than G and the induction hypothesis applies to G'' . By Lemma 3.3, G'' has at most $(1 - \alpha)n$ vertices. As a subgraph of G , G''

3 Odd cycle transversals in graphs without short odd cycles

has no odd cycles shorter than $\varepsilon n = \frac{\varepsilon}{1-\alpha}(1-\alpha)n \geq \frac{\varepsilon}{1-\alpha}|V(G'')|$. Thus, by the induction hypothesis, $\text{oct}(G'') \leq \frac{4(1-\alpha)}{\varepsilon} \ln \left(\frac{15(1-\alpha)}{\varepsilon} \right)$. This means

$$\begin{aligned} \text{oct}(G) &\leq |V'| + \text{oct}(G'') \leq \alpha \frac{4}{\varepsilon} \ln \left(\frac{15}{\varepsilon} \right) + \frac{4(1-\alpha)}{\varepsilon} \ln \left(\frac{15(1-\alpha)}{\varepsilon} \right) \\ &\leq \alpha \frac{4}{\varepsilon} \ln \left(\frac{15}{\varepsilon} \right) + \frac{4(1-\alpha)}{\varepsilon} \ln \left(\frac{15}{\varepsilon} \right) = \frac{4}{\varepsilon} \ln \left(\frac{15}{\varepsilon} \right). \end{aligned}$$

□

To prove Lemma 3.3, we make use of three further Lemmas, the first of which is a useful property of a collection of real numbers.

Lemma 3.4 (from [23]). *Let $\beta > 0$ and $k \in \mathbb{Z}_{\geq 1}$. Further, let $a_0, a_1, \dots, a_k \in \mathbb{R}_{\geq 0}$. If*

$$a_j > \beta \sum_{i=0}^j a_i \quad \forall j \in [k]$$

then

$$\sum_{i=0}^k a_i > a_0 e^{\beta k}.$$

Proof. If $\beta \geq 1$, we have $a_1 > \beta(a_0 + a_1) \geq a_1$. Hence, in this case, the condition is never satisfied and the claim trivially holds.

Assume now $\beta < 1$. We prove the claim by induction on k . For the case $k = 1$, note that $a_1 > \beta(a_1 + a_0)$ implies $a_1 > \frac{\beta}{1-\beta}a_0$. Together with $e^x \leq \frac{1}{1-x}$ for $x < 1$, we obtain $a_0 + a_1 > \left(1 + \frac{\beta}{1-\beta}\right)a_0 = \frac{1}{1-\beta}a_0 \geq a_0 e^{\beta}$. For the inductive step, similar to the base case, we note that $a_k > \beta \sum_{i=0}^k a_i$ implies $a_k > \frac{\beta}{1-\beta} \sum_{i=0}^{k-1} a_i$ and hence

$$\sum_{i=0}^k a_i = a_k + \sum_{i=0}^{k-1} a_i > \left(1 + \frac{\beta}{1-\beta}\right) \sum_{i=0}^{k-1} a_i = \frac{1}{1-\beta} \sum_{i=0}^{k-1} a_i > e^{\beta} (a_0 e^{\beta(k-1)}) = a_0 e^{\beta k}.$$

□

Secondly, we use two observations about odd cycles in graphs, for which we introduce the following notation.

Definition 3.5. *For any two distinct vertices v, w in G let $d_G(v, w)$ denote the minimal number of edges in a v - w -path in G .*

The following elementary observation about shortest odd cycles in graphs was made by Györi et al. [5], but they did not provide a proof. For the sake of completeness, we include the proof here.

Lemma 3.6. *Let C be a shortest odd cycle in G . Then, $d_C(v, w) = d_G(v, w)$ for any two vertices v, w of C .*

Proof. Assume the contrary and let v, w be two vertices of C such that the shortest v - w -path in G has strictly fewer edges than the shortest v - w -path in C . Let P_1, P_2 be the edge sets of the two v - w -paths in C and let Q be the edge set of a shortest v - w -path in G .

Note that for $i \in \{1, 2\}$ we have that $(P_i \cup Q) \setminus (P_i \cap Q)$ is an edge-disjoint union of cycles. Further any cycle in $(P_i \cup Q) \setminus (P_i \cap Q)$ has length at most $|P_i| + |Q| < |C|$, and therefore must be even. This means that $|P_i| + |Q| = |(P_i \cup Q) \setminus (P_i \cap Q)| + 2|P_i \cap Q|$ is even and hence $|C| + 2|Q| = |P_1| + |P_2| + 2|Q|$ is even, a contradiction to $|C|$ being odd. □

Lastly, we use the following observation from Har-Peled and Rahul [24] about odd cycles in graphs in which all vertices are close to some central vertex. They only give a sketch of the proof, which, for the sake of convenience, we develop into a full proof here.

Lemma 3.7. *Let $v \in V$ such that $d_G(v, w) \leq i$ for all $w \in V$. Then, if G contains any odd cycle, it also contains an odd cycle of length at most $2i + 1$.*

Proof. Let C be an odd cycle of length l in G . Let C contain vertices $v_1, \dots, v_l, v_{l+1} = v_1$ where v_i and v_{i+1} are adjacent in C . Then we have $0 = \sum_{i=1}^l (d_G(v, v_i) - d_G(v, v_{i+1}))$ and the right hand side is a sum over an odd number of terms all of which are in $\{-1, 0, +1\}$, thus one of them must be 0. Therefore, two adjacent vertices v_i, v_{i+1} of C have the same distance to v . Let P_1 be the edge set of a shortest v - v_i -path and let P_2 be the edge set of a shortest v - v_{i+1} -path and let $e = \{v_i, v_{i+1}\}$. Since $|P_1| = |P_2|$ and by minimality of P_1 and P_2 , the edge e is in neither P_1 nor P_2 . Consider $G' = (V, E')$ with $E' = ((P_1 \cup P_2) \setminus (P_1 \cap P_2)) \cup \{e\}$. In G' all vertices have even degree and hence it is an edge disjoint union of cycles, further $|E'| = |P_1| + |P_2| - 2|P_1 \cap P_2| + 1 = 2|P_1| - 2|P_1 \cap P_2| + 1$ is odd. Therefore, there is an odd cycle in G' which has length at most $|E'| \leq 2|P_1| + 1 \leq 2i + 1$ and clearly this cycle is also in G . $\square$

We are now ready for the proof of Lemma 3.3. The proof works by starting with a vertex v from a shortest odd cycle and noting that the graph induced by all vertices that have a small distance to v is bipartite. We then remove all vertices with distance exactly i_0 to v for some cleverly chosen i_0 , which splits G into a relatively large bipartite graph and some remainder.

Proof of Lemma 3.3. Let C be a shortest odd cycle in G of length $l \geq \varepsilon n$. Let v be any vertex of C , let $N_i = \{w : d_G(v, w) < i\}$ be all vertices of distance less than i to v and let $\delta_i = \{w : d_G(v, w) = i\}$ be all vertices with distance exactly i to v . Let $k = (l - 1)/2$ and for $i \in [k]$ let H_i be the graph induced by N_i . Note that $N_i = \bigcup_{j=0}^{i-1} \delta_j$.

By Lemma 3.7, for any $i \in [k]$, if H_i contained any odd cycle, it would also contain an odd cycle of length at most $2(i - 1) + 1 \leq 2(k - 1) + 1 < 2k < l$. However, this is impossible since G contains no odd cycles shorter than l . This means H_i is bipartite for any $i \in [k]$.

Now, define $a_i = |\delta_i|$. Since H_i is bipartite, removing any of the δ_i yields a graph in which at most $n - (|N_i| + |\delta_i|) = n - \sum_{j=0}^i a_j$ vertices do not belong to bipartite connected components. Setting $\alpha = 1/n \sum_{j=0}^i a_j$, we see that we can choose $V' = \delta_i$, if we are able to show that $a_i \leq \alpha \frac{4}{\varepsilon} \ln\left(\frac{15}{\varepsilon}\right) = \frac{4}{\varepsilon n} \ln\left(\frac{15}{\varepsilon}\right) \sum_{j=0}^i a_j$. Set $\beta = \frac{4}{\varepsilon n} \ln\left(\frac{15}{\varepsilon}\right)$ and assume that no $i \in [k]$ satisfies this condition, i.e. that $a_i > \beta \sum_{j=0}^i a_j$ for all $i \in [k]$. We will now derive a contradiction.

To this end, let $k' = \lfloor k/3 \rfloor - 1$ and set $\tilde{a}_{k'} = \sum_{i=0}^{k'} a_i$. Note, that v is a vertex of C and C is a shortest odd cycle which means Lemma 3.6 implies $a_j = |\delta_j| \geq 2$ for $j \in [k']$, and therefore we can lower bound $\tilde{a}_{k'}$ as follows

$$\tilde{a}_{k'} = \sum_{j=0}^{k'} a_j \geq 2k' + 1 \geq 2 \left(\frac{k-2}{3} - 1 \right) + 1 = \frac{2k-7}{3} = \frac{(l-1)-7}{3} \geq \frac{\varepsilon n - 8}{3}.$$

Moreover, we have

$$k - (k' + 1) = k - \left\lfloor \frac{k}{3} \right\rfloor \geq \frac{2k}{3} = \frac{l-1}{3} \geq \frac{\varepsilon n - 1}{3}.$$

Using this, Lemma 3.4 applied to $\tilde{a}_{k'}, a_{k'+1}, \dots, a_k$, and since the number of vertices in N_k or δ_k cannot exceed the total number of vertices, we obtain

$$n \geq \sum_{i=0}^k a_i = \tilde{a}_{k'} + \sum_{i=k'+1}^k a_i > \tilde{a}_{k'} e^{\beta(k-k'-1)} \geq \frac{\varepsilon n - 8}{3} e^{\beta(\varepsilon n - 1)/3} \geq \frac{\varepsilon n}{15} e^{\beta(\varepsilon n)/4} = \frac{\varepsilon n}{15} \frac{15}{\varepsilon} = n$$

where we used $\varepsilon n \geq 10$ in the final inequality. Since this is a contradiction, there has to be some $i_0 \in [k]$ such that $a_{i_0} \leq \beta \sum_{j=0}^{i_0} a_j$ and setting $V' = \delta_{i_0}$, $\alpha = 1/n \sum_{j=0}^{i_0} a_j$ satisfies the claimed properties. $\square$

3.2 The weighted case

The weighted case was first considered by Bock et al. in their 2014 conference paper [1], where they stated their version of the result for the weighted case, but postponed the proof to the journal version of the paper, which, unfortunately, was never published. However, they provided us with the unpublished journal version of their paper, for which we are deeply grateful.

The proof presented by Bock et al. [1] follows the outline of the proof by Györi et al. [5] for the unweighted case. However, as Bock et al. note, the proof by Györi et al. is erroneous, and correcting the error introduces additional technical complexities. For this reason, we only take the construction and some crucial ideas for the analysis from their proof and combine them with the proof for the unweighted case given in the previous section. This not only enables us to give an arguably less technical proof but also allows us to improve the given constants and extend the proof to non-negative weights, rather than just positive ones.

For the remainder of the section, let $c : V \rightarrow \mathbb{Z}_{\geq 0}$ denote a weight function that assigns every vertex of G a non-negative integer weight.

The result we will obtain is as follows.

Theorem 3.8. *Let $0 < \varepsilon \leq 1$ and let all odd cycles in G have weight more than $\varepsilon c(V)$. Then $\text{oct}(G) \leq \frac{20}{\varepsilon} \ln\left(\frac{68}{\varepsilon}\right)$.*

As this generalizes the unweighted case, the given bound is also tight up to constant factors. Moreover, the proof also works similarly to that of the unweighted case. Concretely, we rely on a weighted generalization of Lemma 3.3.

Lemma 3.9. *Let $0 < \varepsilon \leq 1$ and let all odd cycles in G have weight more than $\varepsilon c(V) \geq 84$. Then there is a nonempty subset of vertices $V' \subseteq V$ and $\alpha > 0$ such that $|V'| \leq \alpha \frac{20}{\varepsilon} \ln\left(\frac{68}{\varepsilon}\right)$ and removing V' from G yields a graph G' in which the total weight of all vertices which do not belong to bipartite components is at most $(1 - \alpha)c(V)$.*

Using this lemma, we can prove Theorem 3.8.

Proof of Theorem 3.8. If $c(V) = 0$, G is already bipartite, since by assumption all odd cycles have weight strictly more than $\varepsilon c(V) = 0$. If $c(V) > 0$, any cycle has to have positive weight and hence contain at least one vertex of positive weight.

If $\varepsilon c(V) \leq 84$, then $\frac{20}{\varepsilon} \ln\left(\frac{68}{\varepsilon}\right) \geq \frac{20c(V)}{84} \ln\left(\frac{68}{\varepsilon}\right) \geq \frac{20c(V)}{84} \ln(68) \geq c(V)$, so all vertices of positive weight form an odd cycle transversal of the claimed size. If $\varepsilon c(V) \geq 84$, we can show the claim using Lemma 3.9 completely analogous to the unweighted case. $\square$

In order to prove Lemma 3.9, we also need weighted versions of Lemma 3.6 and Lemma 3.7. To this end, we first introduce a notion of distance on weighted graphs.

Definition 3.10. *For any two distinct vertices v, w in G let $d_{G,c}(v, w)$ denote the smallest c -weight of a v - w -path in G (including the weight of v and w), moreover set $d_{G,c}(v, v) = c(v)$.*

Then, we have the following generalization of Lemma 3.6.

Lemma 3.11. *Let C be an odd cycle of minimal weight in G and let $c' = c_C$. Then $d_{G,c}(v, w) = d_{C,c'}(v, w)$ for any two vertices v, w of C . Moreover, for any vertex v of C there is a vertex w of C such that $d_{C,c'}(v, w) \geq \lceil c(C)/2 \rceil$.*

Proof. The proof of the first part works completely analogously to the proof of the unweighted lemma. To prove the second part, let w be a vertex in C which maximizes $d_{C,c'}(v, w)$ and let P_1 be the v - w -path of minimal weight in C . w has two neighboring vertices in C , exactly one of which is not in P_1 . Call this vertex w' and let P_2 be the v - w' -path in C which shares only the vertex v with P_1 . Then $c(C) + c(v) = c(P_1) + c(P_2)$ and by maximality of w we have $c(P_1) \geq c(P_2)$, so $d_{C,c'}(v, w) = c(P_1) \geq (c(P_1) + c(P_2))/2 \geq c(C)/2$ and the claim follows since $d_{C,c'}(v, w)$ is integer. $\square$

Further, we have the following weighted counterpart to Lemma 3.7.

Lemma 3.12. *Let $v \in V$ such that $d_{G,c}(v, w) \leq i$ for all $w \in V$. Then the shortest odd cycle in G has weight at most $4i$.*

Proof. Let C be a shortest odd cycle in G . Let s, t be two vertices which maximize $d_{G,c}(s, t)$. By Lemma 3.11, $d_{G,c}(s, t) \geq \lceil c(C)/2 \rceil$. Therefore, $c(C) \leq 2d_{G,c}(s, t) \leq 2(d_{G,c}(v, s) + d_{G,c}(v, t)) \leq 4i$. $\square$

We are now ready to prove Lemma 3.9. The proof works similarly to the unweighted one, but the presence of weights introduces two significant challenges: First, we need to adapt the notion of a neighborhood to the weighted case in a useful way, and second, we need to relate the weight of a set to its cardinality. The solutions to these challenges presented in the following proof are taken from the unpublished notes of Bock et al. The remainder of the proof follows the outline of our proof for the unweighted case.

Proof of Lemma 3.9. Let C be an odd cycle in G of minimal weight $l \geq \varepsilon c(V)$. Let v be any vertex of C . If $c(v) \geq \varepsilon c(V)/84$, then $c(v) \geq \varepsilon c(V)/(20 \ln(68)) \geq c(V)/(\frac{20}{\varepsilon} \ln(\frac{68}{\varepsilon}))$, so $\alpha = 1/(\frac{20}{\varepsilon} \ln(\frac{68}{\varepsilon}))$ and $V' = \{v\}$ satisfy the requirements.

Assume now that $c(v) \leq \varepsilon c(V)/84$ for all vertices v of C , and let

$$N_i = \{w \in V : d_{G,c}(v, w) < i\},$$

moreover let

$$\delta_i = \begin{cases} \{w \in V : d_{G,c}(v, w) \geq i, w \text{ is adjacent to some } u \in N_i\} & \text{for } i > c(v) \\ \{w \in V : d_{G,c}(v, w) = i\} & \text{for } i = c(v) \end{cases}.$$

Let $k = \lfloor l/4 \rfloor$ and set $I = \{c(v) + 1, c(v) + 2, \dots, k\}$. For $i \in I$ let H_i be the subgraph of G induced by N_i .

By Lemma 3.12, for any $i \in I$ the shortest odd cycle in the graph H_i has weight at most $4(i - 1) \leq 4(k - 1) < 4k \leq l$. Therefore H_i cannot contain any odd cycles, meaning it is bipartite. Moreover, any edge that connects a vertex of H_i with a vertex of $G - H_i$ is incident to a vertex in δ_i , which means removing δ_i from G results in a graph in which all vertices of N_i are in bipartite connected components.

Furthermore, we claim that for $i \in I$ we have $\bigcup_{j=c(v)}^i \delta_j \subseteq N_i \cup \delta_i$. To see this, consider $w \in \bigcup_{j=c(v)}^i \delta_j \setminus N_i$. Then, $d_{G,c}(v, w) \geq i$ and $w \in \delta_j$ for some $c(v) + 1 \leq j \leq i$. Since $N_j \subseteq N_i$, this means $w \in \delta_i$ and the claim follows. In particular, this implies $c(N_i \cup \delta_i) \geq c(\bigcup_{j=c(v)+1}^i \delta_j)$. We now link this cost to the cardinalities of the δ_j . To do this, set $\Delta = \bigcup_{j=c(v)+1}^i \delta_j$ and note that for any $w \in \Delta$, we have $c(w) = |\{j \in \mathbb{Z}_{\geq c(v)+1} : w \in \delta_j\}|$. Using this, we obtain

$$\begin{aligned} \sum_{j=c(v)+1}^i |\delta_j| &= \sum_{j=c(v)+1}^i \sum_{w \in \Delta} |\delta_j \cap \{w\}| = \sum_{w \in \Delta} \sum_{j=c(v)+1}^i |\delta_j \cap \{w\}| \\ &\leq \sum_{w \in \Delta} \sum_{j=c(v)+1}^{\infty} |\delta_j \cap \{w\}| = \sum_{w \in \Delta} |\{i \in \mathbb{Z}_{\geq c(v)+1} : w \in \delta_i\}| \\ &= \sum_{w \in \Delta} c(w) = c(\Delta). \end{aligned}$$

Hence, $\sum_{j=c(v)+1}^i |\delta_j| \leq c(\Delta) \leq c(N_i \cup \delta_i)$.

Now, for $i \in I$, define $a_i = |\delta_i|$. By the considerations above, removing any of the δ_i yields a graph in which vertices of weight at most $c(V) - c(N_i \cup \delta_i)$ do not belong to bipartite connected components. Setting $\alpha = c(N_i \cup \delta_i)/c(V) \geq 1/c(V) \sum_{j=c(v)+1}^i a_j$, we see that we can choose $V' = \delta_i$, if we are able to show that $a_i \leq \frac{20}{\varepsilon c(V)} \ln(\frac{68}{\varepsilon}) \sum_{j=c(v)+1}^i a_j$.

Set $\beta = \frac{20}{\varepsilon c(V)} \ln(\frac{68}{\varepsilon})$ and assume that no $i \in I$ satisfies this condition, i. e. that $a_i > \beta \sum_{j=c(v)+1}^i a_j$ for all $i \in I$.

Let $k' = \lfloor k/5 \rfloor - 1$ and set $\tilde{a}_{k'} = \sum_{i=c(v)+1}^{k'} a_i$. Note, that v is a vertex of C and C is an odd cycle of minimal weight which means Lemma 3.11 implies $|\delta_i| \geq 1$ for $i \in I$, and therefore $\tilde{a}_{k'} \geq k' - c(v)$. We can further lower-bound this in the following way.

$$\begin{aligned} \tilde{a}_{k'} &\geq k' - c(v) = \lfloor k/5 \rfloor - 1 - c(v) \geq \frac{k-9}{5} - c(v) = \frac{4k-36}{20} - c(v) \\ &\geq \frac{(l-3)-36}{20} - c(v) \geq \frac{\varepsilon c(V) - 39}{20} - c(v) \geq \frac{\varepsilon c(V) - 39}{20} - \frac{\varepsilon c(V)}{84} \geq \frac{\varepsilon c(V)}{68} \end{aligned}$$

where we used $c(v) \leq \varepsilon c(V)/84$ in the second to last and $\varepsilon c(V) \geq 84$ in the last inequality.

Moreover, we have

$$k - (k' + 1) = k - \left\lfloor \frac{k}{5} \right\rfloor \geq \frac{4k}{5} \geq \frac{l-3}{5} \geq \frac{\varepsilon c(V) - 3}{5} \geq \frac{\varepsilon c(V)}{20}.$$

Then, using Lemma 3.4 applied to $\tilde{a}_{k'}, a_{k'+1}, \dots, a_k$ we obtain

$$c(N_k \cup \delta_k) \geq \sum_{i=c(v)+1}^k a_i = \tilde{a}_{k'} + \sum_{i=k'+1}^k a_i > \tilde{a}_{k'} e^{\beta(k-k'-1)} \geq \frac{\varepsilon c(V)}{68} e^{\beta \varepsilon c(V)/20} = \frac{\varepsilon c(V)}{68} \frac{68}{\varepsilon} = c(V).$$

Since the weight of $N_k \cup \delta_k$ cannot exceed $c(V)$, this is a contradiction and there has to be some $i_0 \in I$ such that $a_{i_0} \leq \beta \sum_{j=c(v)+1}^{i_0} a_j$ and setting $V' = \delta_{i_0}$, $\alpha = c(N_{i_0} \cup \delta_{i_0})/c(V)$ satisfies the claimed properties. $\square$

3.3 Optimizing the constants

In Theorem 3.1 and Theorem 3.8 the bounds have the form $\frac{a}{\varepsilon} \ln \left(\frac{b}{\varepsilon} \right)$ for two constants a and b . Note that $\frac{a}{\varepsilon} \ln \left(\frac{b}{\varepsilon} \right) / \left(\frac{1}{\varepsilon} \ln \left(\frac{1}{\varepsilon} \right) \right) = a - \frac{a \ln b}{\ln(\varepsilon)} \rightarrow a$ as $\varepsilon \rightarrow 0$, meaning for $\varepsilon \rightarrow 0$ optimizing the constants means minimizing a . Inspecting the proofs of Theorem 3.1 and Theorem 3.8, it is not too difficult to see that for any $\delta > 0$ we can strengthen the bounds to

$$\frac{2+\delta}{\varepsilon} \ln \left(\frac{b_\delta}{\varepsilon} \right)$$

in the unweighted case where b_δ is some constant that only depends on δ and to

$$\frac{4+\delta}{\varepsilon} \ln \left(\frac{b'_\delta}{\varepsilon} \right)$$

in the weighted case where b'_δ is some constant that only depends on δ . For the unweighted case, this was already noted by Komlós [23].

We only outline the choice of b_δ and the changes needed to adapt the proof of the unweighted case here; the weighted case works completely analogously. The key insight is that by requiring $\varepsilon n \geq M$ for a large constant M instead of only $\varepsilon n \geq 5$ in the conditions of Lemma 3.3 and by setting $k' = \lfloor k/N \rfloor - 1$ for some large constant N at the end of the proof of Lemma 3.3, the final chain of inequalities in the proof holds for $a = 2 + \delta$ if b_δ is sufficiently large. In addition, b_δ has to be sufficiently large, so that the claim is trivial for $\varepsilon n < M$.

For small δ , this second condition is the most restrictive on b_δ , meaning one can further improve constants by requiring $\varepsilon n \geq M$ explicitly. Note that the result given by Har-Peled and Rahul [24] is of this form and that our method also yields improved constants for this case.

The limit of $2 + \delta$ in the unweighted, respectively $4 + \delta$ in the weighted case, stems from the fact that k is approximately $\varepsilon n/2$ in the unweighted, respectively $\varepsilon c(V)/4$ in the weighted case. The bound on k can in turn be traced to Lemma 3.7, respectively 3.12. It is not difficult to see that these two lemmas have

the best possible constants. Therefore, to improve constants beyond these bounds, a significantly different approach would be needed.

Lastly, note that when ε is not close to 0, both a and b are relevant, and optimizing them at the same time is less straightforward than just optimizing a , as one has to strike a trade-off between the two constants. Moreover, the best choice of constants depends on the exact value of ε .

4 The stable set polytope of graphs with bounded odd cycle packing number

Previously, in Theorem 2.8, we have seen that the partial convexification $\text{FSTAB}(G)$ with regard to a variable set of constant size coincides with $\text{STAB}(G)$ if $\text{oct}(G)$ is bounded by a constant. In this section, we want to lift this result to the case that only $\text{ocp}(G)$ is bounded, at the expense of not reaching $\text{STAB}(G)$ exactly but having an arbitrarily small integrality gap.

To achieve this, we will use Theorem 3.8 about graphs without short weighted cycles to bound the integrality gap of $\text{FSTAB}(G)$ in terms of $\text{ocp}(G)$. We will subsequently use this bound to derive the desired result.

4.1 Bounding the integrality gap of the fractional stable set polytope

This section is based on notes provided by Weltge and Kober, which they kindly shared.

Theorem 4.1. *There is a function $f(k) = \mathcal{O}(k \ln(k))$ such that for any G and any $c \in \mathbb{Q}^V$ the integrality gap of the stable set problem with weights c satisfies*

$$\frac{\text{OPT}(\text{FSTAB}(G), c)}{\text{OPT}(\text{STAB}(G), c)} \leq f(\text{ocp}(G))$$

if c has at least one positive entry and if c only has non-positive entries, we have

$$\text{OPT}(\text{FSTAB}(G), c) = \text{OPT}(\text{STAB}(G), c) = 0.$$

Note that for G the complete graph and $c(v) = 1$ for all $v \in V$, we have $\text{OPT}(\text{STAB}(G), c) = 1$ and $\text{OPT}(\text{FSTAB}(G), c) \geq |V|/2$ since the all-1/2-vector is feasible. Moreover, $\text{ocp}(G) \geq \lfloor |V|/3 \rfloor$ by partitioning into triangles. This means $\frac{\text{OPT}(\text{FSTAB}(G), c)}{\text{OPT}(\text{STAB}(G), c)} \geq \frac{|V|}{2} = \Omega(\text{ocp}(G))$ demonstrating that the bound is tight up to logarithmic factors.

Proof of Theorem 4.1. The second part of the theorem follows since $\mathbf{0} \in \text{FSTAB}(G)$ and $\mathbf{0} \in \text{STAB}(G)$. Moreover, if G is bipartite, we have $\text{FSTAB}(G) = \text{STAB}(G)$ in which case the claim is trivial. We therefore assume that G is not bipartite.

Since both $\text{FSTAB}(G)$ and $\text{STAB}(G)$ are monotone polyhedra, there is always an optimal solution x^* for either one that satisfies $x_v^* = 0$ for any vertex v with $c_v \leq 0$. Let $\tilde{V} = \{v \in V : c_v > 0\}$ and set $\tilde{G} = G[\tilde{V}]$. Then $\text{FSTAB}(\tilde{G})$ is the face $\text{FSTAB}(G) \cap \{x : x_v = 0 \ \forall v \notin \tilde{V}\}$ of $\text{FSTAB}(G)$ projected onto the $\tilde{V}$ variables. Similarly, $\text{STAB}(\tilde{G})$ is the face $\text{STAB}(G) \cap \{x : x_v = 0 \ \forall v \notin \tilde{V}\}$ of $\text{STAB}(G)$ projected onto the $\tilde{V}$ variables. Because of this, we can assume $c > 0$.

Let $k = \text{ocp}(G)$. For $S \subseteq V$, set

$$\text{LP}(S) = \text{OPT}(\text{FSTAB}(G[S]), c_S) \quad \text{and} \quad \text{IP}(S) = \text{OPT}(\text{STAB}(G[S]), c_S).$$

By greedily choosing c -lightest cycles, we can find node-disjoint odd cycles with vertex sets $C_1, \dots, C_{k'}$ with $k' \leq k$ such that $c(C_i) \leq c(V)/(2k)$ for $i \in [k']$ and such that any odd cycle C in $G' = G[V']$ satisfies $c(C) \geq c(V)/(2k)$ where $V' = V \setminus \bigcup_{i=1}^{k'} C_i$.

Using this, and the fact that the all-1/2 vector is feasible for $\text{LP}(V')$ we get

$$2\text{LP}(V') \geq c(V') \geq c(V) - \sum_{i=1}^{k'} c(C_i) \geq c(V) - \frac{k'c(V)}{2k} \geq c(V) - \frac{c(V)}{2} = \frac{c(V)}{2}.$$

Therefore, we have

$$\text{LP}(V) \leq \text{LP}(V') + \sum_{i=1}^{k'} \text{LP}(V(C_i)) \leq \text{LP}(V') + \frac{k'c(V)}{2k} \leq \text{LP}(V') + \frac{c(V)}{2} \leq 3\text{LP}(V')$$

where the first inequality holds since $\text{LP}(V) \leq \text{LP}(V \setminus S) + \text{LP}(S)$ for any $S \subseteq V$. Observing that $\text{IP}(V') \leq \text{IP}(V)$ since $V' \subseteq V$, we can use this to estimate the integrality gap with respect to G in terms of G' as follows

$$\frac{\text{LP}(V)}{\text{IP}(V)} \leq \frac{\text{LP}(V)}{\text{IP}(V')} \leq \frac{3\text{LP}(V')}{\text{IP}(V')}.$$

To bound the integrality gap with respect to G' , let T be a minimal-size odd cycle transversal in G' . Then, we have

$$\frac{\text{LP}(V')}{\text{IP}(V')} \leq \frac{\text{LP}(V')}{\text{IP}(V' \setminus T)} = \frac{\text{LP}(V')}{\text{LP}(V' \setminus T)} \leq \frac{\text{LP}(V' \setminus T) + \text{LP}(T)}{\text{LP}(V' \setminus T)} = 1 + \frac{\text{LP}(T)}{\text{LP}(V' \setminus T)}, \quad (4.1)$$

where the first equality holds since $G[V' \setminus T]$ is bipartite. On the other hand, we also have

$$\frac{\text{LP}(V')}{\text{IP}(V')} \leq \frac{\text{LP}(V')}{\text{IP}(T)} \leq |T| \frac{\text{LP}(V')}{\text{LP}(T)} \leq |T| \frac{\text{LP}(V' \setminus T) + \text{LP}(T)}{\text{LP}(T)} \leq |T| \left(1 + \frac{\text{LP}(V' \setminus T)}{\text{LP}(T)} \right), \quad (4.2)$$

where the second inequality holds, since $\text{LP}(T) \leq |T| \max\{c(v) : v \in T\} \leq |T| \text{IP}(T)$. Combining equations (4.1) and (4.2), we get

$$\frac{\text{LP}(V')}{\text{IP}(V')} \leq \min \left\{ 1 + \frac{\text{LP}(T)}{\text{LP}(V' \setminus T)}, |T| \left(1 + \frac{\text{LP}(V' \setminus T)}{\text{LP}(T)} \right) \right\} \leq 2|T|,$$

where we used that $|T| \geq 1$ and the fact that the two fractions are inverse to each other, meaning that one of them must be at least 1.

By Theorem 3.8, T has size at most $|T| \leq 40k \ln(136k)$, allowing us to conclude

$$\frac{\text{LP}(V)}{\text{IP}(V)} \leq \frac{3\text{LP}(V')}{\text{IP}(V')} \leq 6|T| \leq 240k \ln(136k).$$

□

Corollary 4.2. *There is a function $g(k) = \mathcal{O}(k \ln(k))$ such that for any non-bipartite graph G and any $c \in \mathbb{Q}^V$ it holds that*

$$\text{OPT}(\text{STAB}(G), c) \geq \frac{c(V)}{g(\text{ocp}(G))}.$$

Proof. Since the all-1/2 vector is in $\text{FSTAB}(G)$, we have $\text{OPT}(\text{FSTAB}(G), c) \geq c(V)/2$. Combining this with Theorem 4.1 yields the claim. □

4.2 Strength of the Balas-Ceria-Cornuéjols hierarchy

As observed before, we can equivalently phrase Theorem 2.8 by stating that for any vector $c \in \mathbb{R}^V$ the integrality gap $\frac{\text{OPT}(\text{FSTAB}(G)^{(T),c})}{\text{OPT}(\text{STAB}(G),c)} = 1$ for any graph G and any odd cycle transversal T in G . We now want to prove a generalization of this statement for graphs with bounded $\text{ocp}(G)$, which will come at the cost of having an integrality gap of $1 + \varepsilon$ instead of 1.

Note that for any graph G we have $\text{ocp}(G) \leq \text{oct}(G)$. Moreover, Reed [6] gave a family of graphs, attributed to Lovász and Schrijver, whose odd cycle packing number is much smaller than the odd cycle transversal number. Concretely, he presents a family of graphs which satisfy $\text{ocp}(G) = 1$ and $\text{oct}(G) = \Omega(\sqrt{|V|})$. This means requiring a bounded odd cycle packing number is a much weaker assumption than requiring a bounded odd cycle transversal number.

Under this weaker assumption, we will obtain the following result.

Theorem 4.3. *For any graph $G = (V, E)$ with $\text{ocp}(G) = k$, any node weights $c \in \mathbb{Q}^V$ with at least one positive entry, and any $\varepsilon > 0$, there is a subset $T \subseteq V$ of size $|T| = \mathcal{O}\left(\frac{k^2 \ln(k)}{\varepsilon} \ln\left(\frac{k}{\varepsilon}\right)\right)$ such that*

$$\frac{\text{OPT}(\text{FSTAB}(G)^{(T)}, c)}{\text{OPT}(\text{STAB}(G), c)} \leq 1 + \varepsilon.$$

Proof. Since both $\text{FSTAB}(G)^{(T)}$ and $\text{STAB}(G)$ are monotone polyhedra, we can again assume $c \geq \mathbf{0}$.

Let $k = \text{ocp}(G)$, let g be the function from Corollary 4.2 and set $\varepsilon' = \varepsilon/g(k)$. As in the proof of Theorem 4.1, by greedily choosing c -lightest cycles, we can find node-disjoint odd cycles with vertex sets $C_1, \dots, C_{k'}$ with $k' \leq k$ such that $c(C_i) \leq \varepsilon' c(V)/k$ for $i \in [k']$ and such that any odd cycle C in $G' = G[V \setminus V']$ satisfies $c(C) \geq \varepsilon' c(V)/k$ where $V' = \bigcup_{i=1}^{k'} C_i$. By Theorem 3.8 there is an odd cycle transversal $T \subseteq V \setminus V'$ in G' with

$$|T| = \mathcal{O}\left(\frac{k}{\varepsilon'} \ln\left(\frac{k}{\varepsilon'}\right)\right) = \mathcal{O}\left(\frac{k^2 \ln(k)}{\varepsilon} \ln\left(\frac{k^2 \ln(k)}{\varepsilon}\right)\right) = \mathcal{O}\left(\frac{k^2 \ln(k)}{\varepsilon} \ln\left(\frac{k}{\varepsilon}\right)\right).$$

Moreover,

$$c(V') = \sum_{i=1}^{k'} c(C_i) \leq k' \frac{\varepsilon' c(V)}{k} \leq \varepsilon' c(V).$$

By using Lemma 2.10 in the first and Corollary 4.2 in the final inequality, we conclude

$$\frac{\text{OPT}(\text{FSTAB}(G)^{(T)}, c)}{\text{OPT}(\text{STAB}(G), c)} \leq 1 + \frac{c(V')}{\text{OPT}(\text{STAB}(G), c)} \leq 1 + \frac{\varepsilon' c(V)}{\text{OPT}(\text{STAB}(G), c)} \leq 1 + \varepsilon' g(k) = 1 + \varepsilon.$$

□

By pairing this result with Theorem 2.4, we see that for any fixed $\varepsilon > 0$, any constant k , and any graph G with $\text{ocp}(G) \leq k$, we can determine the optimal value of the the stable set problem on G up to a factor of $1 + \varepsilon$ in polynomial time. This result is not new: Bock et al. [1] showed the stronger result that there is a PTAS for the stable set problem on a more general class of graphs, and, as mentioned before, Fiorini et al. [4] recently showed that the problem even is solvable in polynomial time. However, both of these results are based on combinatorial algorithms, while ours is purely polyhedral and therefore of independent interest.

5 Perfect formulations

In the previous chapters, we have introduced the stable set polytope and its standard LP-relaxation, the fractional stable set polytope. We have seen that these two polytopes do not coincide in general and have explored a way to bridge the difference between the two, or at least approximately do so, based on integer programming hierarchies. In this section, we will explore an alternative approach. Instead of applying integer programming hierarchies to the fractional stable set polytope, we will consider LP-relaxations that are stronger than the fractional stable set polytope. Since the stable set problem is NP-hard, we cannot expect to find a complete set of well-behaved inequalities that suffice to reach the stable set polytope exactly in all cases. However, we will see that in some special cases, this is indeed possible. We refer to such an exact linear description of the stable set polytope as a *perfect formulation*.

5.1 Perfect graphs

Recall that the fractional stable set polytope of a graph $G = (V, E)$ is given by

$$\text{FSTAB}(G) = \{x \in [0, 1]^V : x_v + x_w \leq 1 \text{ for all } \{v, w\} \in E\}.$$

It is easy to see that these constraints are valid for all points in the stable set polytope, since a stable set cannot contain two adjacent points. Viewing edges as cliques of size two, we can strengthen this in a natural way to the following polytope

$$\text{QSTAB}(G) = \left\{ x \in [0, 1]^V : \sum_{v \in Q} x_v \leq 1 \text{ for all cliques } Q \subseteq V \right\}.$$

In the literature, this is also often referred to as the *fractional stable set polytope* instead of $\text{FSTAB}(G)$. The constraints being used are called *clique-constraints*. Clearly, all constraints corresponding to inclusion-wise not maximal cliques are dominated by the constraints corresponding to inclusion-wise maximal cliques. Moreover, the constraints corresponding to inclusion-wise maximal cliques are all facet defining for $\text{STAB}(G)$ [25].

It is immediately clear that $\text{STAB}(G) \subseteq \text{QSTAB}(G) \subseteq \text{FSTAB}(G)$. In general, both of these inclusions can be strict. Graphs for which we have $\text{STAB}(G) = \text{QSTAB}(G)$, i.e. for which the clique constraints yield a perfect formulation, are known as *perfect graphs*. Interestingly, there are also combinatorial definitions of perfect graphs.

Theorem 5.1. *For any graph G , the following are equivalent.*

- (i) $\text{STAB}(G) = \text{QSTAB}(G)$.
- (ii) *For every induced subgraph, the maximal size of a clique equals its chromatic number.*
- (iii) *No induced subgraph of G is an odd cycle of length at least five or the complement of one.*
- (iv) *The complement graph of G is perfect.*

Proof. The original definition is (ii). A proof of the equivalence of (i) can be found in [26]. The equivalence of (iii) was shown by Chudnovsky et al. [27] and is known as the *strong perfect graph theorem*. The equivalence of (iv) was shown by Lovász [28] and is known as the *perfect graph theorem*. $\square$

Furthermore, more than a hundred classes of graphs are known to be perfect. These include, but are not limited to: Bipartite graphs, line graphs of bipartite graphs, interval graphs, comparability graphs, chordal graphs, parity graphs, Gallai graphs, and Meyniel graphs. An extensive overview including known inclusions between the classes has been compiled by Hougardy [29].

It is also important to mention that the separation problem of $\text{QSTAB}(G)$ for graphs is NP-hard, meaning optimizing over $\text{QSTAB}(G)$ in general is NP-hard as well. However, it turns out that it is possible to optimize over $\text{QSTAB}(G)$ in polynomial time for any perfect graph G . This was shown by Grötschel et al. [16] based on work by Lovász [30]. In particular, this means that the stable set problem for perfect graphs is solvable in polynomial time. Remarkably, this polyhedral approach is the only known way to obtain a polynomial-time algorithm, in contrast to our previously presented results, where combinatorial alternatives exist.

5.2 t -perfect graphs

Recall that for a graph $G = (V, E)$, $\text{FSTAB}(G)$ equals $\text{STAB}(G)$ if and only if G is bipartite, or, equivalently, if G does not contain odd cycles. This means the minimal examples for which $\text{FSTAB}(G) \neq \text{STAB}(G)$ are the odd cycles. It is therefore natural to examine which inequalities need to be added to obtain an exact description of the stable set polytope of odd cycles.

Given an odd cycle C of length $2k + 1$, it is easy to see that any stable set can contain at most k vertices of C . This idea yields the polytope given by

$$t\text{STAB}(G) = \left\{ x \in [0, 1]^V : \begin{array}{l} x_v + x_w \leq 1 \text{ for all } \{v, w\} \in E, \\ \sum_{v \in V(C)} x_v \leq \frac{|V(C)| - 1}{2} \text{ for all odd cycles } C \text{ in } G \end{array} \right\}.$$

The constraints based on odd cycles with chords are implied by those of chordless odd cycles and the edge constraints; therefore, the constraints based on odd cycles are usually restricted to chordless odd cycles, which are also known as odd holes. Consequently, these constraints are known as *odd-hole constraints*.

Clearly, for an odd cycle C , the all-1/2 vector is feasible for $\text{FSTAB}(C)$ but not for $t\text{STAB}(C)$, meaning this is a strict improvement upon the fractional stable set polytope. Actually, it turns out that adding this single constraint yields $\text{STAB}(C)$ exactly. However, in general, we have $\text{STAB}(G) \subseteq t\text{STAB}(G) \subseteq \text{FSTAB}(G)$ with both inclusions possibly being strict.

Graphs for which $t\text{STAB}(G) = \text{STAB}(G)$ are called *t -perfect graphs* (where t stands for *trou*, French for hole [26]). t -perfect graphs include series-parallel graphs, nearly bipartite planar graphs, and strongly t -perfect graphs [26]. Moreover, odd cycles are t -perfect, which follows from the more general result that any graph G with $\text{oct}(G) = 1$ is t -perfect. This was shown by Fonlupt and Uhry [31] (albeit using a slightly different terminology).

Optimizing over $t\text{STAB}(G)$ is possible in polynomial time since the separation problem comes down to finding a shortest odd cycle in an edge-weighted graph, see [26] for a detailed explanation. This means the stable set problem for t -perfect graphs can be solved in polynomial time. In particular, similar to Theorem 2.8, this provides a polyhedral algorithm to solve the stable set problem for graphs with odd cycle transversal number one, which, unlike the previously presented algorithm, does not require the introduction of additional variables.

The odd hole constraints can also be derived from the edge constraints as a CG cut: Adding the constraints corresponding to all edges in a chordless odd cycle C yields

$$\sum_{v \in V(C)} 2x_v \leq |V(C)|,$$

dividing by two and rounding down the right-hand side yields the odd hole constraint. Therefore, the CG closure of the fractional stable set polytope $\text{FSTAB}(G)'$ is contained in $t\text{STAB}(G)$. Interestingly, the converse can also be shown, see e. g. [32]. We therefore have the following result.

Theorem 5.2. $\text{FSTAB}(G)' = t\text{STAB}(G)$ for any graph G .

Because of this, graphs for which the i -th CG closure $\text{FSTAB}(G)^{(i)}$ equals $\text{STAB}(G)$ are sometimes called t^i -perfect [33].

5.3 h -perfect graphs

To derive $\text{QSTAB}(G)$ from $\text{FSTAB}(G)$, we replaced the edge constraints by the more general clique constraints. We can do the same starting with $t\text{STAB}(G)$ to obtain

$$h\text{STAB}(G) = \left\{ x \in [0, 1]^V : \sum_{v \in Q} x_v \leq 1 \text{ for all cliques } Q \subseteq V, \right. \\ \left. \sum_{v \in V(C)} x_v \leq \frac{|V(C)| - 1}{2} \text{ for all odd cycles } C \text{ in } G \right\}.$$

Clearly, this refines $t\text{STAB}(G)$, so we have $\text{STAB}(G) \subseteq h\text{STAB}(G) \subseteq t\text{STAB}(G) \subseteq \text{FSTAB}(G)$. Graphs for which $h\text{STAB}(G) = \text{STAB}(G)$ are called h -perfect graphs. Few examples of graphs that are h -perfect but not t -perfect are known. One construction for h -perfect graphs was given by Sbihi and Uhry [34] who showed that graphs obtained by substituting bipartite graphs for edges of a series-parallel graph are h -perfect.

It was first shown by Grötschel et al. [35] that the stable set problem can be solved in polynomial time for h -perfect graphs. Later, Lovász and Schrijver [36] showed an even stronger result when introducing their hierarchy of relaxations for 0/1-polytopes.

5.4 Valid and facet-defining inequalities

One property of the stable set polytope that we have implicitly used in all of the above is the relation of the stable set polytope of a graph $G = (V, E)$ to the stable set polytopes of its induced subgraphs. To illustrate this, consider a subset $V' \subseteq V$ and the induced subgraph $G' = G[V']$. First, we can observe that $\text{STAB}(G')$ is the projection of the face $\text{STAB}(G) \cap \{x : x_v = 0 \text{ for all } v \notin V'\}$ of $\text{STAB}(G)$ onto the V' -variables. From this, we can deduce that any facet-defining inequality $\sum_{v \in V'} a_v x_v \leq \beta$ for $\text{STAB}(G')$ with $a \in \mathbb{R}^{V'}$, $\beta \in \mathbb{R}$ also is a valid inequality for $\text{STAB}(G)$. This fact underlies all of the constraints from before that are based on subgraphs. Although these inequalities are not always facet defining for $\text{STAB}(G)$, Nemhauser and Trotter [37] and, independently, Padberg [38] have shown that any such inequality can be lifted to a facet-defining inequality in the sense that there is $a' \in \mathbb{R}^{V \setminus V'}$ such that $\sum_{v \in V'} a_v x_v + \sum_{v \notin V'} a'_v x_v \leq \beta$ is facet defining for $\text{STAB}(G)$. Moreover, Padberg [25] has shown that for the odd-hole constraints introduced in Section 5.2, a' can be chosen to have non-zero entries only in the neighborhood of V' . The devised lifting procedures are algorithmic; however, they require solving NP-hard problems as a subroutine. Moreover, in general, not all facet-defining inequalities of $\text{STAB}(G)$ can be obtained in this way. For example, Trotter [39] gave a family of graphs whose stable set polytopes have facets that cannot be obtained from any strict subgraph.

Moreover, all of the previously presented linear relaxations of the stable set polytope can be viewed within a unified framework: Given a family of graphs $\mathcal{F}$ such that the stability number $\alpha(H)$, i. e. the maximal size of a stable set in H , is known for all $H \in \mathcal{F}$, we obtain a linear relaxation of the stable set polytope $\text{STAB}(G)$ given by

$$\left\{ x \in [0, 1]^V : \sum_{v \in S} x_v \leq \alpha(H) \text{ for all } S \subseteq V \text{ and } H \in \mathcal{F} \text{ such that } G[S] \text{ is isomorphic to } H \right\}.$$

$\text{FSTAB}(G)$ is obtained by setting $\mathcal{F}$ equal to the two-vertex graph with a single edge, $\text{QSTAB}(G)$ is obtained by setting $\mathcal{F}$ to the set of cliques, $\text{tSTAB}(G)$ is obtained by setting $\mathcal{F}$ to the set of odd cycles and the two-vertex graph with a single edge and lastly $\text{hSTAB}(G)$ is obtained by setting $\mathcal{F}$ to the union of the set of cliques with the set of odd cycles.

Constraints of this type are called *rank constraints* and they have the following property.

Proposition 5.3. *Let $G = (V, E)$ be a graph and let $a \in \{0, 1\}^V, \beta \in \mathbb{R}$ be such that $a^\top x \leq \beta$ is facet-defining for $\text{STAB}(G)$. Then $a^\top x \leq \beta$ is the rank constraint associated to the subgraph induced by $\text{supp}(a) = \{v \in V : a_v \neq 0\}$, i. e. $\beta = \alpha(G[\text{supp}(a)])$.*

Proof. Since a maximal stable set in $G[\text{supp}(a)]$ also is a stable set in G and $a^\top x \leq \beta$ is valid for $\text{STAB}(G)$, we get $\beta \geq \alpha(G[\text{supp}(a)])$. On the other hand, $a^\top x \leq \alpha(G[\text{supp}(a)])$ is clearly valid for $\text{STAB}(G)$. Since $a^\top x \leq \beta$ is facet defining, this means $\beta \leq \alpha(G[\text{supp}(a)])$. $\square$

Note that the converse is not true, i. e. there are rank constraints that are not facet defining. The maybe most straightforward way to see this is to consider any rank constraint that is not an edge constraint for a bipartite graph. We will see further examples of this shortly. However, under certain conditions, rank constraints are facet-defining. For example, Chvátal [12] provided such a sufficient condition.

Besides rank constraints, many other families of valid inequalities have been found, and for many of them, sufficient conditions for an inequality to be facet-defining are known. It is impossible to provide a complete overview of all known families, but we will try to mention some of the most prominent ones as well as explain the ones relevant to the upcoming sections in more detail. An interesting family of valid inequalities are the *orthogonality constraints*, which include clique constraints and are also closely related to a semidefinite relaxation of the stable set polytope known as the Lovász-Theta-Body [16, 30]. The separation problem for them is polynomial-time solvable. Moreover, they stand out because they are not based on subgraphs but rather on orthogonal representations of the full graph. Another family of constraints that numerous papers have been written on are the so-called *clique family constraints* [40, 41, 42]. There are also the *cycle of cycles constraints*, which have also been shown to be polynomial time separable [43].

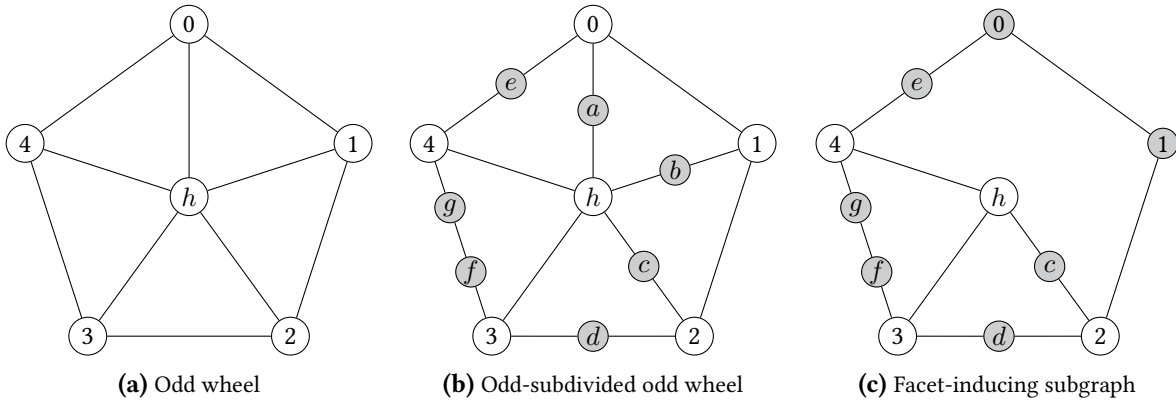


Figure 5.1 Example of an odd wheel and an odd-subdivided odd wheel as well as a facet-inducing subgraph for the subdivided wheel.

The next family is based on so-called *odd wheels*. An odd wheel of size k is a graph that consists of an odd cycle of length k , and an additional vertex that is adjacent to all vertices of the cycle. The cycle is called *rim* and the additional vertex is called *hub*. In Figure 5.1a, an odd wheel of length five is depicted.

Given a graph $G = (V, E)$ and $C \subseteq V, h \in V$ such that $G[C \cup \{h\}]$ is an odd wheel with hub h , we obtain the following valid inequality for $\text{STAB}(G)$

$$\sum_{v \in C} x_v + \frac{|C| - 1}{2} x_h \leq \frac{|C| - 1}{2}. \quad (5.1)$$

To see that this is valid for $\text{STAB}(G)$, consider any stable set $S \subseteq V$ in G . If $h \in S$, we have $C \cap S = \emptyset$, since h is adjacent to all vertices in C . Therefore, the inequality is valid in this case. If $h \notin S$, the inequality is valid, because the odd-hole inequality for $G[C]$ is. Note that this constraint strictly dominates the rank constraint associated with the odd wheel graph. Constraints of this type are called *odd wheel constraints*, and sometimes a more general class of constraints is also referred to as odd wheel constraints. These were introduced by Cheng and Cunningham [44], and they are based on subdivisions of odd wheels. Given an odd wheel with hub h and the odd cycle having vertices $\{v_0, \dots, v_{k-1}\}$, a subdivision replaces the edges $\{h, v_i\}$ by paths P_i for $i \in \{0, \dots, k-1\}$ and the edges $\{v_i, v_{i+1}\}$ by paths $P_{i,i+1}$ for $i \in \{0, \dots, k-1\}$ where the indices are taken modulo k . If the cycles formed by $P_i, P_{i,i+1}$ and P_{i+1} are odd for all i , we say that the resulting graph is an *odd-subdivided odd wheel*. An example of an odd-subdivided odd wheel can be seen in Figure 5.1b. Given such a graph and defining $E = \{v_i : P_i \text{ has even length}\}$, $O = \{v_i : P_i \text{ has odd length}\}$ and S as the set of internal vertices of all paths, Cheng and Cunningham derived the following two valid inequalities for the stable set polytope

$$\begin{aligned} \frac{k-1}{2}x_h + \sum_{i=0}^{k-1} x_i + \sum_{v \in E} x_v + \sum_{v \in S} x_v &\leq \frac{k + |E| + |S| - 1}{2}, \\ \frac{k+1}{2}x_h + \sum_{i=0}^{k-1} x_i + \sum_{v \in O} x_v + \sum_{v \in S} x_v &\leq \frac{k + |O| + |S|}{2}. \end{aligned} \quad (5.2)$$

These are derived as a CG cut of $t\text{STAB}(G)$ and therefore have CG rank two by Theorem 5.2. Note that (5.2) coincides with (5.1) in the case of an odd wheel without subdivision.

In the example from Figure 5.1b, we have $E = \{0, 1, 2\}$, $O = \{3, 4\}$ and $S = \{a, \dots, g\}$ is the set of gray vertices. We get the inequalities

$$\begin{aligned} 2x_h + 2x_0 + 2x_1 + 2x_2 + x_3 + x_4 + x_a + x_b + x_c + x_d + x_e + x_f + x_g &\leq 7, \\ 3x_h + x_0 + x_1 + x_2 + 2x_3 + 2x_4 + x_a + x_b + x_c + x_d + x_e + x_f + x_g &\leq 7. \end{aligned}$$

Interestingly, neither of these two inequalities is facet-defining for the stable set polytope of this graph, but there are strict subgraphs that are also odd-subdivided odd wheels, which do yield facet-defining inequalities. One such subgraph is depicted in Figure 5.1c, and the corresponding facet-defining inequality is

$$x_h + 2x_2 + x_3 + x_4 + x_0 + x_1 + x_c + x_d + x_e + x_f + x_g \leq 5.$$

Cheng and Cunningham gave necessary and sufficient conditions for these inequalities to be facet-defining for the stable set polytope of the underlying odd-subdivided odd wheel. In particular, constraints of the form 5.1 are always facet-defining for the corresponding odd wheel. Moreover, they also gave a generalization of these inequalities, which are based on subdivisions of graphs that arise from an odd wheel by replacing the hub with a clique.

Two related families of inequalities are the *web* and *antiweb constraints* introduced by Trotter [39]. They are the rank constraints of so-called webs and their complements. A web on n vertices with parameter k where $n \geq 2k$, denoted by $W(n, k)$ is a graph whose vertices can be labeled $\{v_0, \dots, v_{n-1}\}$ and v_i is adjacent to exactly $v_{i+k}, v_{i+k+1}, \dots, v_{i+n-k}$ where indices are taken modulo n . The web $W(8, 3)$ is depicted in Figure 5.2a and the web $W(7, 3)$ is depicted in Figure 5.2b. Note that $W(n, 1)$ is the complete graph on n vertices K_n , $W(2n+1, n)$ is an odd hole and $W(2n+1, 2)$ is the complement of an odd-hole.

We will now derive the stability number of a web, which will allow us to write out the corresponding rank inequality explicitly. To this end, let S be any stable set in $W(n, k)$. Due to symmetry, we can assume $v_0 \in S$ without loss of generality. Note that v_0 is adjacent to $v_k, v_{k+1}, \dots, v_{n-k}$ which means

$$S \subseteq \{v_{n-k+1}, v_{n-k+2}, \dots, v_{n-1}, v_0, v_1, \dots, v_{k-1}\}.$$

Note further that the edges

$$\{v_{n-k+1}, v_1\}, \{v_{n-k+2}, v_2\}, \dots, \{v_{n-1}, v_{k-1}\}$$

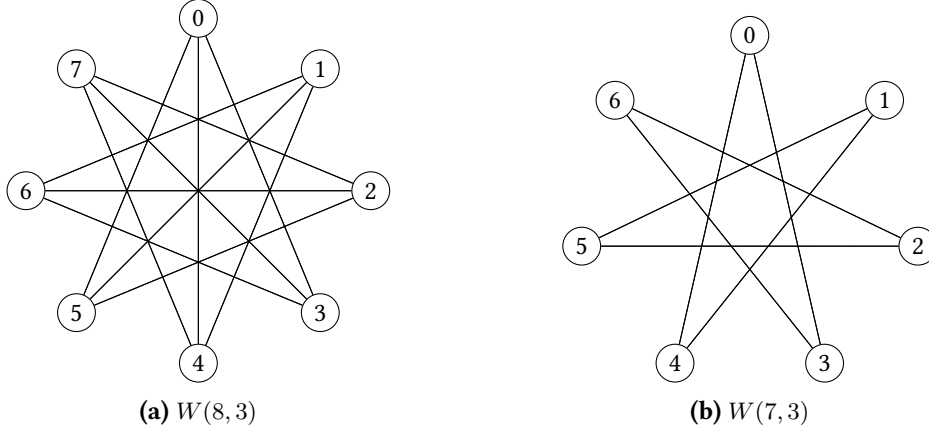


Figure 5.2 Examples of webs.

are present in $W(n, k)$. Therefore, S contains at most one of each of these pairs of vertices, which yields

$$|S| \leq 1 + |\{v_{n-k+1}, v_{n-k+2}, \dots, v_{n-1}, v_1, v_2, \dots, v_{k-1}\}|/2 = 1 + (k-1) = k.$$

Conversely, $\{v_0, v_1, \dots, v_{k-1}\}$ is a stable set of size k in $W(n, k)$. Combining these two facts, we get $\alpha(W(n, k)) = k$, which means the resulting rank inequality is

$$\sum_{i=0}^{n-1} x_i \leq k,$$

where the variable x_i corresponds to v_i for $i \in \{0, \dots, n-1\}$. Trotter showed that this constraint is facet-defining for $\text{STAB}(W(n, k))$ if and only if n and k are coprime. This means for a web $W(n, k)$ which is a subgraph of a graph G to yield a facet-defining web constraint for $\text{STAB}(G)$, n and k have to be coprime.

For the sake of completeness, we also briefly mention the corresponding anti-web constraint that is valid for $\text{STAB}(\overline{W(n, k)})$, where $\overline{G}$ denotes the complement graph of G . It is given by

$$\sum_{i=0}^{n-1} x_i \leq \left\lfloor \frac{n}{k} \right\rfloor.$$

Cheng and de Vries [45] later introduced a common generalization of odd wheel and antiweb inequalities, which they called *antiweb-wheel inequalities*.

Lastly, we would like to introduce two methods for generating new constraints from known families. The first one is based on an observation by Cheng and Cunningham [44].

Proposition 5.4. *Let $G = (V, E)$ be a graph on $V = \{v_0, v_1, \dots, v_n\}$ such that v_0 and v_1 are not adjacent. Let further $a \in \mathbb{R}^V, \beta \in \mathbb{R}$ such that $\sum_{v \in V} x_v \leq \beta$ is valid for $\text{STAB}(G)$. If $G' = (V', E')$ arises from G by identifying v_0 and v_1 , i. e. $V' = V \setminus \{v_0\}$ and $E' = \{e \in E : v_0 \notin e\} \cup \{\{v_1, v_i\} : \{v_0, v_i\} \in E\}$, then $(a_{v_0} + a_{v_1})x_{v_0} + \sum_{v \in V' \setminus \{v_1\}} a_v \leq \beta$ is valid for $\text{STAB}(G')$.*

This proposition enables us to obtain valid inequalities for $\text{STAB}(G)$ not only from subgraphs of G belonging to one of the aforementioned families but also from the larger class of subgraphs which arise from these families by sequential identification of vertices. Cheng and Cunningham [44] as well as Cheng and de Vries [45] applied this to extend the families of valid inequalities that we described above.

The second way to derive new inequalities is based on the join of two graphs.

Definition 5.5. *Let $G = (V_1, E_1)$ and $H = (V_2, E_2)$ be two graphs on disjoint vertex sets. Then*

$$G \vee H = (V_1 \cup V_2, E_1 \cup E_2 \cup \{\{u, v\} : u \in V_1, v \in V_2\})$$

is called the join of G and H .

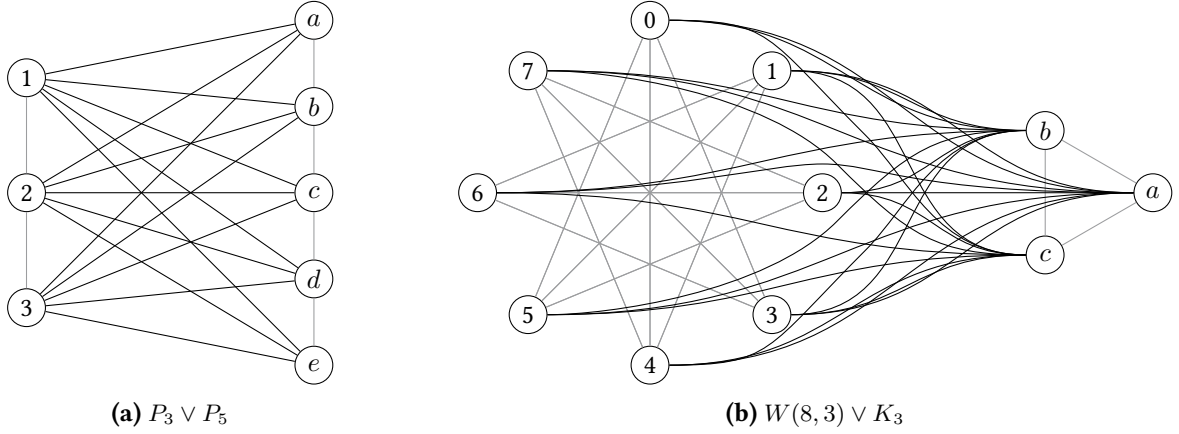


Figure 5.3 Examples of joins. Edges of the underlying graphs are drawn in gray, joining edges are drawn in black.

In Figure 5.3a, the join of the paths on three and five vertices is depicted, and in Figure 5.3b, the join of $W(8, 3)$ and the complete graph on three vertices is depicted. Using this, we obtain another possibility for generating new constraints, which was mentioned without proof by Shepherd [2] and studied in a more general setting by Chvátal [12].

Proposition 5.6. *Given graphs $G_i = (V_i, E_i)$ for $i \in [k]$ on disjoint vertex sets, the inequality*

$$\sum_{i=1}^k \left(\frac{1}{\alpha(G_i)} \sum_{v \in V_i} x_v \right) \leq 1$$

is valid for $\text{STAB}(G_1 \vee G_2 \vee \dots \vee G_k)$. It is called the join of the rank inequalities corresponding to the G_i .

Proof. Let S be a stable set in $G_1 \vee G_2 \vee \dots \vee G_k$. Clearly $S \subseteq V_i$ for some $i \in [k]$. This means $\sum_{j=1}^k \left(\frac{1}{\alpha(G_j)} \sum_{v \in V_j} (\chi(S))_v \right) = \frac{1}{\alpha(G_i)} \sum_{v \in V_i} (\chi(S))_v$ which is at most one by the rank constraint for G_i . $\square$

Applying this to $P_3 \vee P_5$ from Figure 5.3a, we get the inequality

$$\frac{1}{2} (x_1 + x_2 + x_3) + \frac{1}{3} (x_a + x_b + x_c + x_d + x_e) \leq 1$$

and applying it to $W(8, 3) \vee K_3$ from Figure 5.3b, we get

$$\left(\frac{1}{3} \sum_{i=0}^7 x_i \right) + (x_a + x_b + x_c) \leq 1.$$

Moreover, applying this to odd cycles and a single vertex yields the odd wheel constraints.

5.5 Line graphs, quasi-line graphs and their complements

Given a graph $G = (V, E)$ its *line graph* is given by $L(G) = (E, \{\{e, f\} : e, f \in E, |e \cap f| = 1\})$, i.e. the vertices of $L(G)$ correspond to the edges of G and two edges $e, f \in E$ are adjacent in $L(G)$ if they have a common endpoint. The name line graph was first used by Harary and Norman [46]. Since stable sets in $L(G)$ are precisely matchings in G , the linear description of the matching polytope given by Edmonds in his seminal paper [47] also yields a linear description of $\text{STAB}(L(G))$. Note that $L(G)$ has the property that the neighborhood of any of its vertices can be partitioned into two cliques, one coming from each endpoint of the associated edge in G . This property is the defining property of *quasi-line graphs*. The problem of determining a description of the stable set polytope of quasi-line graphs was first tackled by

Ben Rebea [48] in his doctoral thesis in 1981. Unfortunately, he died shortly after, before being able to fix some errors in his claimed solution. This work was built on by Oriolo [40], who formulated a conjecture for the linear description of quasi-line graphs, which he named Ben Rebea's Conjecture in his memory. This conjecture was finally proven by Eisenbrand et al. in 2008 [49]. For the sake of brevity and since this result will not be relevant in the upcoming sections, we will not reproduce it here and refer to the original publications instead.

At this point, we also feel obliged to mention *claw-free graphs*, which are graphs that do not have a $K_{1,3}$ as an induced subgraph, i. e. no vertex has three pairwise non-adjacent neighbors. Claw-free graphs have been extensively studied, see [50] for a survey, and their stable set polytope was considered as early as 1981 [51]. In 1988, Grötschel, Lovász, and Schrijver observed that "in spite of considerable efforts, no decent system of inequalities describing $\text{STAB}(G)$ is known" and this remains this way to this day, nearly four decades later. We mention claw-free graphs here, since they include quasi-line graphs as a proper subclass, and quasi-line graphs are probably the most important subclass of claw-free graphs for which a decent linear description is known.

The stable set polytope of complements of line graphs and quasi-line graphs was studied by Shepherd [2]. He studied these graphs under the name *near bipartite*, due to the following equivalent definition.

Proposition 5.7. *A graph $G = (V, E)$ is the complement of a quasi-line graph if and only if $G[V \setminus N(v)]$ is bipartite for any $v \in V$.*

Proof. For $v \in V$ we use the notation $N_G(v)$, respectively $N_{\overline{G}}(v)$, to refer to the neighborhood of v in G , respectively in $\overline{G}$. Let G be the complement of a quasi-line graph and let $v \in V$. Then $N_{\overline{G}}(v)$ can be partitioned into two cliques in G . Noting that $N_G(v) = V \setminus N_{\overline{G}}(v)$ and that the complement of a clique is a stable set, we see that $G[V \setminus N_G(v)]$ can be partitioned into two stable sets. This is just a rephrasing of $G[V \setminus N_G(v)]$ being bipartite. Conversely, let G be such that $G[V \setminus N_G(v)]$ is bipartite for any $v \in V$ and let $u \in V$. Symmetrically, we see that $N_{\overline{G}}(u) = V \setminus N_G(u)$, which means the complement graph of $\overline{G}[N_{\overline{G}}(u)]$ is bipartite. By the correspondence of cliques and stable sets, this means $\overline{G}[N_{\overline{G}}(u)]$ can be partitioned into two cliques. $\square$

To state a linear description of the stable set polytope for complements of quasi line graphs, Shepherd introduced the notion of a *prime web*: A web $W(n, k)$ is said to be prime, if $n = 2k + 1 + l$ for some $l \in \mathbb{Z}_{\geq 0}$ and n and k are coprime¹. Using this, he was able to prove the following complete linear description of the stable set polytope of complements of quasi-line graphs by leveraging a matrix-theoretic result due to Lehman [52].

Theorem 5.8. *Let $G = (V, E)$ be the complement of a quasi-line graph. Then any facet-defining inequality for $\text{STAB}(G)$ is either of the form $x_v \geq 0$ for some $v \in V$, or it is the join of clique and prime web inequalities, where join is in the sense of Proposition 5.6.*

Shepherd also specialized this result to the complement of line graphs, in which case the additional structure allows for only very specific joins of clique and prime web inequalities. Although this result does not transfer directly to the odd-cycle constrained setting that we consider, we will see some conclusions that can be drawn from it in the following section.

5.6 Complements of quasi-line graphs with small odd cycle transversals

In this section, we will consider the complement graphs of quasi-line graphs with a small odd cycle transversal number. As mentioned before, for any bipartite graph G , that is any graph with $\text{oct}(G) = 0$, the fractional stable set polytope and the stable set polytope coincide. Moreover, as we saw previously, Fonlupt and Uhry showed that any graph G with $\text{oct}(G) = 1$ is t -perfect. We summarize this in the following proposition.

¹Note that the terminology used by Shepherd is slightly different. We use the definition of Trotter [39], which switches webs and antiwebs compared to the convention that Shepherd uses.

Proposition 5.9. *Let G be a graph. Then it holds that*

- (i) $\text{FSTAB}(G) = \text{STAB}(G)$ if $\text{oct}(G) = 0$,
- (ii) $t\text{STAB}(G) = \text{STAB}(G)$ if $\text{oct}(G) = 1$.

Already for graphs G with $\text{oct}(G) = 2$, to the best of our knowledge, there is no perfect formulation of $\text{STAB}(G)$ known. However, it turns out that Shepherds' result from Theorem 5.8 can be specialized in a nice way to the cases where additionally $\text{oct}(G) = 2$ or 3 holds. To do this, we establish some tools to pin down the relevant inequalities. The first one is the following simple observation.

Lemma 5.10. *For any graph $G = (V, E)$ it holds that $\text{oct}(G) \geq |V| - 2\alpha(G)$*

Proof. Let T be a minimal odd cycle transversal. Then $G[V \setminus T]$ is bipartite, which means it has at most $2\alpha(G)$ many vertices, since either side of a bipartition is a stable set. We conclude $|V| - \text{oct}(G) = |V| - |T| \leq 2\alpha$. $\square$

From this, we can deduce the odd cycle transversal number of the graphs that are relevant for Theorem 5.8.

Lemma 5.11. *$\text{oct}(W(2k+1+l, k)) = l+1$ for any $k \geq 1, l \geq 0$ and $\text{oct}(K_n) = n-2$ for any $n \geq 2$.*

Proof. As discussed before, $\alpha(W(2k+1+l, k)) = k$. Using Lemma 5.10, we get $\text{oct}(W(2k+1+l, k)) \geq (2k+1+l) - 2k = l+1$. Conversely, the set $\{v_0, \dots, v_l\}$ is an odd cycle transversal of size $l+1$. To see this, observe that $\{v_{l+1}, \dots, v_{l+k}\}$ and $\{v_{l+k+1}, \dots, v_{l+2k}\}$ are stable sets in $W(2k+1+l, k)$ which means the graph induced by their union is bipartite. Together, we obtain $\text{oct}(W(2k+1+l, k)) = l+1$, as claimed.

For the complete graph, by Lemma 5.10, $\text{oct}(K_n) \geq n-2$. Since any graph on two vertices is bipartite, any set of $n-2$ vertices is an odd cycle transversal, and equality holds. $\square$

Moreover, we have a simple relation between the odd cycle transversal number of the join of a graph and the odd cycle transversal numbers of its constituent graphs.

Lemma 5.12. *For any two graphs G_1, G_2 we have $\text{oct}(G_1 \vee G_2) \geq \text{oct}(G_1) + \text{oct}(G_2)$. Moreover any odd cycle transversal in $G_1 \vee G_2$ yields an odd cycle transversal in G_i when intersected with the vertices of G_i for $i \in \{1, 2\}$.*

Proof. This holds since $G \vee H$ contains G and H as disjoint subgraphs. $\square$

Furthermore, complete graphs play a special role in joins and in the join of rank inequalities.

Lemma 5.13. *Given graphs G_i for $i \in [k]$ and integers $n_1, \dots, n_l \in \mathbb{Z}_{\geq 1}$, the join of the rank inequalities corresponding to $G_1, \dots, G_k, K_{n_1}, \dots, K_{n_l}$ is equal to the join of the rank inequalities corresponding to $G_1, \dots, G_k, K$ where $K = K_{n_1} \vee \dots \vee K_{n_l}$.*

Since $K_{n_1} \vee \dots \vee K_{n_l}$ is again a complete graph, we can deduce from this lemma that, without loss of generality, we only need to consider at most one clique in the join of rank inequalities.

Lastly, we can limit the number of graphs that we need to consider the join of by using the following lemma.

Lemma 5.14. *Let $W_i = W(2k_i+1+l_i, k_i)$ be two prime webs for $i \in \{1, 2\}$ and let G be the graph on a single vertex. Then $\text{oct}(W_1 \vee W_2 \vee G) \geq 4$.*

Proof. Assume the contrary and let T be an odd cycle transversal of size 3 in $W_1 \vee W_2 \vee G$. Note that each W_i contains a vertex of degree at least 2. Let v_1 be such a vertex in W_1 and let u_1, u_2 denote two of its neighbors in W_1 . Similarly, let v_2 be such a vertex in W_2 and let w_1, w_2 denote two of its neighbors in W_2 . Lastly, let v^* be the vertex of G . The situation is depicted in Figure 5.4. Then $\{v_1, u_1, v_2, w_1, v^*\}$ and $\{v_1, u_2, v_2, w_2, v^*\}$ induce cliques of size 5 in $W_1 \vee W_2 \vee G$. By Lemma 5.11, these cliques have odd cycle

transversal number 3. Therefore, T has to be a subset of both these sets, hence also their intersection, which means $T = \{v_1, v_2, v^*\}$. Now, we see that T intersects W_1 in only one vertex, meaning $l_1 = 0$ due to Lemma 5.11. This means that W_1 is an odd hole. However, given that W_1 is an odd hole, we can construct an odd cycle that does not intersect T by replacing v_1 by w_1 . Therefore, we have arrived at a contradiction. $\square$

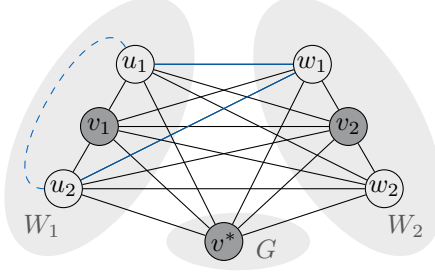


Figure 5.4 Illustration of the situation in the proof of Lemma 5.14. The vertices belonging to T are filled in gray. The graphs that the vertices belong to are indicated by the gray ellipses. The contradicting cycle found at the end of the proof is highlighted in blue.

We are now ready to put together all the pieces and prove the following theorem.

Theorem 5.15. *Let $G = (V, E)$ be the complement of a quasi-line graph with $\text{oct}(G) \leq 2$. Then $\text{STAB}(G)$ is completely determined by constraints of the following types*

- (i) $0 \leq x_v \leq 1$ for $v \in V$,
- (ii) edge constraints,
- (iii) odd hole constraints,
- (iv) odd wheel constraints,
- (v) web constraints corresponding to webs $W(4k, 2k - 1)$ for some $k \geq 1$.

Proof. By Theorem 5.8, all facet-defining inequalities are either non-negativity constraints or joins of clique and prime web inequalities. Using Lemma 5.13 and Lemma 5.14, we see that we only need to consider joins of at most two inequalities. Moreover, by Lemma 5.11, the only relevant cliques are K_1, K_2, K_3 and K_4 and the only relevant webs are of the form $W(2k + 1 + 0, k)$ and $W(2k + 1 + 1, k)$ for some $k \geq 1$ with the two arguments being coprime. Note that $2k + 1 + 0$ and k are coprime for any k and that $2k + 1 + 1$ and k are coprime if and only if k is odd. This means we need to consider webs $W(2k + 1, k)$ which we will call W_0^k and of the form $W(4k, 2k - 1)$ which we will call W_1^k . Note that W_0^k is an odd hole of length $2k + 1$.

The constraints given by the join of a single clique, which is equivalent to just that clique constraint, yield $x_v \leq 1$ for some $v \in V$ for K_1 , an edge constraint for K_2 , an odd hole constraint for K_3 , and an odd wheel constraint for K_4 . The constraints given by the join of a single web inequality yield an odd hole constraint for W_0^k and a constraint of type (v) for W_1^k .

The join of two clique inequalities is again a clique inequality by Lemma 5.13. Therefore, it remains to consider the join of two web inequalities and the join of a web and a clique inequality. By Lemma 5.12 and Lemma 5.11, we can rule out the join of some W_0^k and K_4 as well as the join of some W_1^k and K_3, K_4, W_0^l, W_1^l .

The join of some W_0^k and K_3 has a K_5 as a subgraph, since W_0^k has at least one edge, and therefore has odd cycle transversal number at least 3. The join of some W_0^k and K_2 as well as the join of some W_0^k and some W_0^l has odd cycle transversal number at least 3, since after deleting one vertex from the W_0^k there is still an edge left in it and clearly there is also an edge in the other graph which induce a K_4 in the join.

Consider now the join of some W_1^k with a K_1 . For $k = 1$, this is a K_4 , which we have already considered. Therefore, we can assume $k \geq 2$. By Lemma 5.12, any odd cycle transversal has to contain two vertices of W_1^k . However, the three vertex-disjoint edges $\{v_0, v_{2k-1}\}, \{v_1, v_{2k}\}, \{v_2, v_{2k+1}\}$ of W_1^k induce three copies of K_3 in the join with K_1 and any odd cycle transversal that only contains two vertices of W_1^k can intersect at most two of them. Thus, the join of a W_1^k with a K_1 has odd cycle transversal number at least 3. By the same argument, the join of W_1^k with K_2 has odd cycle transversal number at least 3.

Lastly, the join of a K_1 and a W_0^k yields an odd wheel constraint. $\square$

Recall that all of the mentioned families of inequalities are facet-defining for the stable set polytope of the respective graphs, e. g. the odd wheel constraints are facet-defining for the underlying odd wheel. Furthermore, it is easy to check that all of the graphs underlying these constraints are themselves complements of quasi-line graphs. Therefore, the given description is minimal in the sense that none of the five families can be omitted.

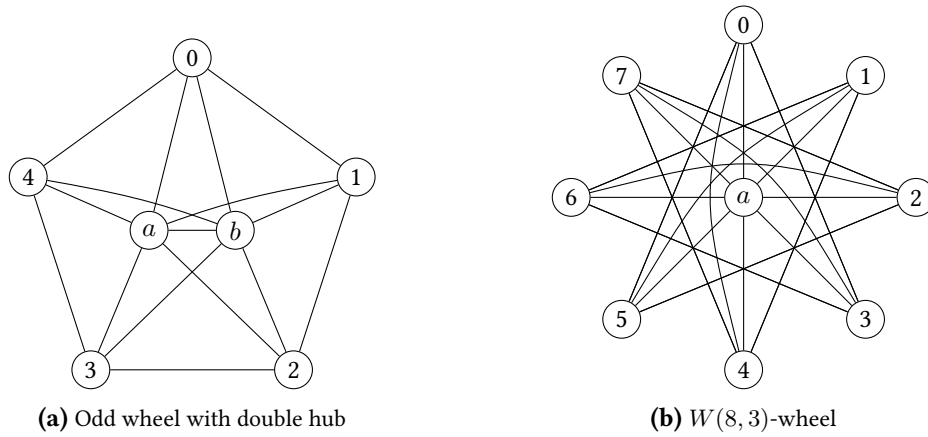


Figure 5.5 Examples of the graphs relevant for Theorem 5.16.

A corresponding theorem for the case $\text{oct}(G) = 3$ can also be obtained. To this end, we need to introduce two new types of graphs and the related inequalities. The first one is the join of an odd cycle with a K_2 , which we will call an *odd wheel with double hub* and the respective constraint, which we will call *odd wheel with double hub constraint*. Analogous to odd wheels, we will call the cycle the *rim* and the K_2 the *hub*. The second one is the join of a $W(n, k)$ with a single vertex, which we will call a $W(n, k)$ -wheel and the corresponding constraint, which we will call $W(n, k)$ -wheel constraint or, less specific a *web wheel constraint*. We will call the single vertex the *hub* of the web wheel. In Figure 5.5a, an odd wheel with double hub is depicted, and the corresponding inequality is

$$2x_a + 2x_b + \sum_{i=0}^4 x_i \leq 2. \quad (5.3)$$

In Figure 5.5b an $W(11, 4)$ -wheel is depicted and the corresponding inequality is

$$3x_a + \sum_{i=0}^7 x_i \leq 3.$$

With these preliminaries in place, we can state the result.

Theorem 5.16. *Let $G = (V, E)$ be the complement of a quasi-line graph with $\text{oct}(G) \leq 3$. Then $\text{STAB}(G)$ is completely determined by constraints (i) to (v) from Theorem 5.15 and the following types*

- (vi) *web constraints corresponding to webs $W(6k + 1, 3k - 1)$ or $W(6k - 1, 3k - 2)$ for some $k \geq 1$,*
- (vii) *odd wheel with double hub constraints,*

(viii) $W(4k, 2k - 1)$ -wheel constraints for some $k \geq 1$.

Proof. Again, by Theorem 5.8, all facet-defining inequalities are either non-negativity constraints or joins of clique and prime web inequalities, and using Lemma 5.13 and Lemma 5.14, we see that we only need to consider joins of at most two inequalities. Moreover, by Lemma 5.11, the only relevant cliques are K_1, K_2, K_3, K_4 and K_5 and the only relevant webs are of the form W_0^k, W_1^k , where we reuse the notation of the proof of Theorem 5.15, and $W(2k + 1 + 2, k)$ for some $k \geq 1$ with the two arguments being coprime. Note that $2k + 3$ and k are coprime if and only if k is not divisible by three. This yields the webs $W_{2,1}^k = W(6k + 1, 3k - 1)$ and $W_{2,2}^k = W(6k - 1, 3k - 2)$ for $k \geq 1$.

K_5 can be seen as an odd cycle of length 3 with a double hub, which means the corresponding inequality is accounted for. The web constraints corresponding to $W_{2,1}^k$ and $W_{2,2}^k$ are precisely (vi).

Thus, it remains to consider the joins of two web inequalities and the join of a web and a clique inequality. The joins with odd cycle transversal number at most two have been handled already in the proof of Theorem 5.8 and remain relevant. By Lemma 5.12 and Lemma 5.11, we can rule out the join of a W_0^k and a K_5 as well as the join of a W_1^k with a K_4, K_5 or W_1^l and the join of a $W_{2,1}^k$ or $W_{2,2}^k$ with a K_3, K_4, K_5 or a $W_0^l, W_1^l, W_{2,1}^l, W_{2,2}^l$.

Consider the join of a W_0^k with a K_3, K_4 or W_0^l . By Lemma 5.12, any odd cycle transversal in the join has to intersect either graph in at least one vertex. However, after deleting one vertex from either graph, there is still an edge in either graph present. These two edges induce a K_4 in the join, which means at least two further vertices need to be in an odd cycle transversal. This means the odd cycle transversal number of such a join is at least 4. Similarly, after deleting two vertices from a W_1^k , there is an edge left, meaning the join with a K_2, K_3 or W_0^k has odd cycle transversal number at least 4. Now consider the join of a $W_{2,1}^k$ with a K_1 . By Lemma 5.12, any odd cycle transversal of the join has to contain three vertices of $W_{2,1}^k$. Note that $W_{2,1}^k$ is a 4-regular graph on at least 7 vertices, which means after deleting three vertices, there certainly is an edge left. This edge induces a K_3 when joined with the K_1 , meaning the odd cycle transversal number of the join of a $W_{2,1}^k$ with a K_1 is at least 4. For $W_{2,2}^k$ when $k \geq 2$ the same argument applies and $W_{2,2}^1$ is a K_5 . Clearly, this also means the join of $W_{2,1}^k$ or $W_{2,2}^k$ with a K_2 has odd cycle transversal number at least 4.

Lastly, the join of a W_k^1 with a K_1 yields constraints of type (viii) and the join of a W_0^k with a K_2 yields constraints of type (vii). $\square$

Note that the work of Cheng and Cunningham [44] implies that constraints of type (vi) and (vii) cannot be omitted (we did not present the latter result, it is part of their work on webs whose hub has been replaced by a clique which we mentioned only in passing). Moreover, since webs of the form $W(4k, 2k - 1)$ are facet-defining for their stable set polytope, there are $4k$ stable sets whose characteristic vectors are affinely independent. Lifting these to the graph with the additional vertex present in a $W(4k, 2k - 1)$ -wheel and adding the stable set consisting only of that additional node yields a collection of $4k + 1$ stable sets in the $W(4k, 2k - 1)$ -wheel whose characteristic vectors are affinely independent. This shows that the $W(4k, 2k - 1)$ -wheel constraints are also facet-defining for their underlying graphs and, in particular, (viii) cannot be omitted. Furthermore, as was the case for Theorem 5.15, it is easy to check that all of the graphs underlying the given families of constraints are themselves complements of quasi-line graphs. This means also the characterization given by Theorem 5.16 is best possible.

One interesting class of graphs, which are complements of quasi-line graphs, which is connected to odd cycle transversals is based on the following definition that we introduce.

Definition 5.17. Let $G = (V, E)$ be a graph, $T \subseteq V$ a set of vertices. T is said to be a fully connected odd cycle transversal if

- (i) T is an odd cycle transversal and
- (ii) any $t \in T$ is adjacent to all $v \in V \setminus T$.

In Figure 5.6, a graph with a fully connected odd cycle transversal is shown. Now, the reason for considering fully connected odd cycle transversals is given in the following proposition.

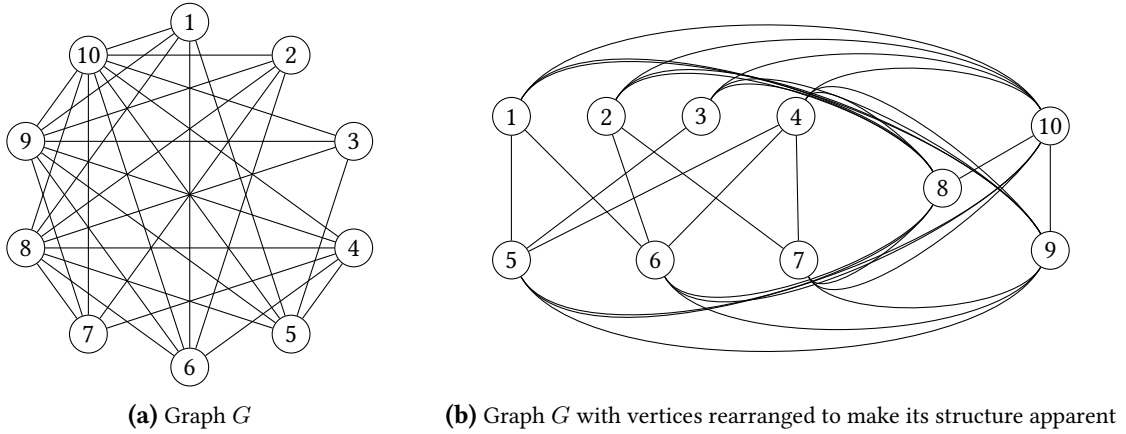


Figure 5.6 Example of a graph with a fully connected odd cycle transversal $T = \{8, 9, 10\}$ of size 3.

Proposition 5.18. *If a graph G has a fully connected odd cycle transversal T and $G[T]$ is the complement of a quasi-line graph, then G also is the complement of a quasi-line graph.*

Proof. Let $G = (V, E)$ be a graph with T a fully connected odd cycle transversal. According to Proposition 5.7, we need to show that $G[V \setminus N(v)]$ is bipartite for any $v \in V$. Let $v \in V \setminus T$. Because of condition (ii), $T \subseteq N(v)$ which means $G[V \setminus N(v)]$ is a subgraph of the bipartite graph $G[V \setminus T]$. Let now $v \in T$, then $V \setminus T \subseteq N(v)$ because of condition (ii). Let $H = G[T]$, then this means that $G[V \setminus N_G(v)]$ is equal to $H[T \setminus N_H(v)]$, which is bipartite since H is assumed to be the complement of a quasi-line graph. $\square$

Note that, in particular, any graph on at most three vertices is the complement of a quasi-line graph. This yields the following corollary.

Corollary 5.19. *Let G be a graph with a fully connected odd cycle transversal of size 2, respectively 3. Then, G is the complement of a quasi-line graph and Theorem 5.15, respectively Theorem 5.16, is applicable.*

Observe that due to Theorem 2.8, for graphs G with odd cycle transversal number $r \in \{0, \dots, 3\}$, any of the hierarchies mentioned in Section 2.3 reaches $\text{STAB}(G)$ exactly after r iterates. In particular graphs G as in Theorem 5.15, respectively 5.16, have this property. We finish this section by showing an analogue of this result for the CG closure.

Theorem 5.20. *Let $r \in \{0, \dots, 3\}$ and let G be the complement of a quasi-line graph with $\text{oct}(G) = r$. Then the CG rank of $\text{FSTAB}(G)$ is at most r . Moreover, this bound is attained for each $r \in \{0, \dots, 3\}$.*

Proof of Theorem 5.20. In the **case** $r = 0$, $\text{STAB}(G) = \text{FSTAB}(G) = \text{STAB}(G)^{(0)}$. In the **case** $r = 1$, the CG rank of $\text{FSTAB}(G)$ is at most one by Theorem 5.2 and the fact that graphs with odd cycle transversal number 1 are t -perfect. Moreover, it is exactly one for odd cycles, which are complements of quasi-line graphs with odd cycle transversal number 1. It remains to show the cases $r \in \{2, 3\}$.

We first handle the **case** $r = 2$. According to 5.15, we only have to consider the families of constraints (i) to (v). The constraints (i) and (ii) trivially have CG rank 0. (iii) have CG rank 1 according to 5.2. (iv) were derived by Cheng and Cunningham as a CG cut based on non-negativity, edge, and odd-hole constraints and consequently have CG rank at most 2. To see that the web constraints of type (v) have CG rank at most 2, first consider a general web $W(n, m)$ and a shortest odd-cycle with vertices $v_{i_1}, v_{i_2}, \dots, v_{i_l}$ in this web. By symmetry of the web, for $j \in \{0, \dots, n-1\}$ there is an odd cycle with vertices $v_{i_1+j}, v_{i_2+j}, \dots, v_{i_l+j}$ where the indices are taken modulo n . Note that every vertex appears in exactly l of these cycles. Thus, summing over the corresponding odd-hole constraints, we get

$$l \sum_{i=0}^{n-1} x_i \leq n \frac{l-1}{2}$$

from which we can derive

$$\sum_{i=0}^{n-1} x_i \leq \left\lfloor \frac{n(l-1)}{2l} \right\rfloor$$

as a CG cut. Note that $W(n, m)$ contains an odd cycle of length at most $2m+1$, since no induced subgraph on more than $2\alpha(W(n, m)) = 2m$ vertices can be bipartite. Therefore, $l \leq 2m+1$. Applying this to the webs $W(4k, 2k-1)$ of (v), we obtain the CG cut

$$\sum_{i=0}^{4k-1} x_i \leq \left\lfloor \frac{4k(2(2k-1) + 1 - 1)}{2(2(2k-1) + 1)} \right\rfloor = \left\lfloor \frac{4k}{4k-1}(2k-1) \right\rfloor = 2k-1,$$

which is exactly the relevant web constraint. This means these web constraints can be derived from odd-hole constraints as a CG cut, so their CG rank is at most 2. We have now shown that any inequality defining $\text{STAB}(G)$ for such graphs has CG rank at most two with respect to $\text{FSTAB}(G)$; therefore, $\text{FSTAB}(G)$ has CG rank at most two. Since $\frac{1}{3}\mathbf{1} \in t\text{STAB}(K_4) \setminus \text{STAB}(K_4) = \text{FSTAB}(K_4)' \setminus \text{STAB}(K_4)$, the CG rank is at least two in some cases.

We now turn our attention to the **case** $\mathbf{r} = \mathbf{3}$. Using the same approach that worked for the webs in (v), we can show that the web constraints in (vi) have CG rank at most 2 as well. We now examine the odd wheel with double hub constraints (vii). Let a and b denote the hub vertices and let $1, \dots, 2k+1$ denote the rim vertices of such a graph. Consider the set of 4-cliques that are formed by the two hub-vertices and two adjacent vertices of the rim. Then, summing over all of these constraints, we obtain

$$2 \sum_{i=1}^{2k+1} x_i + (2k+1)x_a + (2k+1)x_b \leq 2k+1$$

adding $-x_a \leq 0$ and $-x_b \leq 0$, dividing by two, and rounding down the right-hand side yields

$$\sum_{i=1}^{2k+1} x_i + kx_a + kx_b \leq k$$

which is the desired inequality. Therefore, these inequalities have CG rank at most one more than the 4-clique inequalities, which can be viewed as odd-wheel inequalities and therefore have CG rank at most 2. To derive the web-wheel inequalities in (viii) as a CG cut, we adapt the derivation of the web inequalities presented before. Let the hub be denoted by h and let the vertices of the web be denoted by $0, \dots, n-1$. Note that for each cycle used in the derivation of the web constraint, there is a corresponding wheel with h as the hub. Summing over the corresponding wheel inequalities yields

$$l \sum_{i=0}^{n-1} x_i + n \frac{l-1}{2} x_h \leq n \frac{l-1}{2}$$

where again l denotes the length of the odd cycles used. Dividing by l yields

$$\sum_{v \in V} x_v + \frac{n(l-1)}{2l} x_h \leq \frac{n(l-1)}{2l}.$$

By adding a non-negative multiple of $-x_h \leq 0$ and rounding down the right side, we now obtain the relevant constraint. This means we can get the web-wheel constraints (viii) by a single step of the CG procedure from non-negativity and wheel constraints, so these inequalities have CG rank at most 3.

To show that there are examples of inequalities that have CG rank at least 3, consider an odd-cycle of length 5 with a double hub as is depicted in Figure 5.5a. Call the graph G and label the hub vertices by a and b and the rim vertices by $0, \dots, 4$. We assume for contradiction that we could derive the corresponding constraint by two steps of the CG procedure, which, according to 5.2, is equivalent to deriving it by a

single step applied to $t\text{STAB}(G)$. Since the constraint is facet-defining for $\text{STAB}(G)$, it cannot be implied by other CG cuts but has to be derived as a CG cut itself. This means there are non-negative multipliers for the constraints defining $t\text{STAB}(G)$ such that the left-hand side becomes the left-hand side of (5.3) and the right-hand side is less than 3. The relevant constraint can be grouped into a few groups: The edge constraints for the edges $\{i, i+1\}$ for $i \in \{0, \dots, 3\}$ and $\{4, 0\}$, the edge constraint for $\{a, b\}$, the edge constraints for the edges $\{h, i\}$ where $h \in \{a, b\}$ and $i \in \{0, \dots, 4\}$, the odd hole consisting of $\{0, \dots, 4\}$, the triangles containing both $\{a\}$ and $\{b\}$ and the triangles containing exactly one of a and b . By symmetry of G , the multipliers for each group can be assumed to be the same for all members of this group. Moreover, with this assumption the edge constraints for the edges $\{i, i+1\}$ for $i \in \{0, \dots, 3\}$ and $\{4, 0\}$ are dominated by the odd hole constraint corresponding to the odd hole consisting of $\{0, \dots, 4\}$ and can be ignored. Let $\lambda_1, \dots, \lambda_5$ denote the multipliers corresponding to the remaining five groups in the order they were mentioned in. Then, the assumption that they give the desired inequality as a CG cut can be summarized in the following system of equations and inequalities

$$\lambda_1 + 5\lambda_2 + 5\lambda_4 + 5\lambda_5 = 2 \quad (5.4)$$

$$2\lambda_2 + \lambda_3 + \lambda_4 + 4\lambda_5 = 1 \quad (5.5)$$

$$\lambda_1 + 10\lambda_2 + 2\lambda_3 + 5\lambda_4 + 10\lambda_5 < 3 \quad (5.6)$$

$$\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5 \geq 0 \quad (5.7)$$

where the first equality comes from the coefficients of the hub vertices, the second equality comes from the coefficients of the rim vertices, and (5.6) comes from the condition on the right-hand side. The constants in front of the λ_i are determined by the coefficients of the hub vertices for (5.4), of the rim vertices for (5.5), and the right-hand side for (5.6). For example, the term $5\lambda_2$ in (5.4) stems from the fact that there are 5 constraints of the second type, i. e. edges connecting a hub vertex to a rim vertex, for each hub vertex, and the hub vertices have coefficient 1 in these inequalities. We can see that there is no solution, since adding $\frac{2}{3}$ times (5.4), $\frac{5}{3}$ times (5.5) and $\frac{1}{3}$, respectively $\frac{10}{3}$ and $\frac{1}{3}$, constraint (5.7) for λ_1 , respectively λ_2 and λ_3 , yields

$$\lambda_1 + 10\lambda_2 + 2\lambda_3 + 5\lambda_4 + 10\lambda_5 \geq 3,$$

in contradiction to 5.6. We conclude that the constraint corresponding to G has CG rank at least 3. $\square$

6 Graphs with odd cycle transversal number 2

In the previous section, we have seen that there is a concise description of the stable set polytope of graphs with odd cycle transversal number 2 that are the complement of a quasi-line graph. We have also mentioned that, to the best of our knowledge, such a description is not known for general graphs with odd cycle transversal number 2. In this section, we want to investigate the role of the constraints that are relevant in the case of complements of quasi-line graphs in the case of general graphs.

The first easy observation is that in the general case, the set of constraints (i) to (v) does not suffice to define the stable set polytope. To see this, observe that any subdivision of an odd wheel graph still has odd cycle transversal number 2, and recall that the odd-subdivided odd wheel inequalities found by Cheng and Cunningham [44] are facet-defining under certain conditions.

Moreover, one can find examples where facets are relevant that are neither of the types (i) to (v) nor odd-subdivided odd wheel constraints. We could not attribute these remaining facets to any class of facets known in the literature, nor could we find any common (combinatorial) structure underlying them that would allow us to explain them as part of a novel family. Therefore, we will not attempt to give a complete description of the stable set polytope of graphs with odd cycle transversal number 2; instead, we will examine computationally the number and share of facets that belong to (and do not belong to) the previously mentioned types.

6.1 Computational setup

Since a complete enumeration of all graphs and subsequent computation of their stable set polytopes is computationally infeasible for even a small number of nodes on the hardware available to us, we instead define a suitable random graph model from which we will sample and analyze graphs.

We use a random graph model inspired by the classic Erdős-Rényi-Gilbert model introduced by Gilbert [53] and jointly named after Erdős and Rényi because of their related seminal work in [54, 55, 56]. Many variations of these models have been explored. See Bollobas [57] for general background and detailed results. One such model is the random graph $G(H; p)$, which is defined for any fixed graph H . This is a random graph on the vertex set of H such that any edge contained in H is included with probability p and any edge not contained in H is not included. Note that we can sample from this distribution by drawing a value q_e uniformly at random from $[0, 1]$, independently for all edges $e \in E(H)$ and including e in the resulting graph if and only if $q_e \leq p$.

Since we want to experiment with random graphs having odd cycle transversal number 2, it is natural to consider $H = K_{n,m} \vee K_2$ for some combination of $n, m \in \mathbb{Z}_{\geq 1}$ since any graph with odd cycle transversal number 2 is a subgraph of such a graph. We denote $G(K_{n,m} \vee K_2; p)$ by $G(n, m; p)$.

Having sampled a random graph $G = (V, E)$, we first compute the set $X = \{\chi(S) \in \{0, 1\}^V : S \text{ is a stable set in } G\}$ by total enumeration¹. Subsequently, we compute all facet-defining inequalities of $\text{STAB}(G) = \text{conv}\{X\}$ using the Julia package of polymake [59], which is developed as part of the OSCAR project [60, 61].

Moreover, given an inequality of the form $\sum_{v \in V} a_v x_v \leq b$, we implemented a classifying algorithm that checks which of the classes (i) to (v) (or odd-subdivided odd wheel constraints) it belongs to. Our approach roughly works as follows: First, we compute $S = \text{supp}(a)$ and the induced graph $G[S]$. Then, we check if the graph structure of $G[S]$ falls into any of the categories underlying the relevant families

¹We are aware that the runtime of computing all stable sets could probably be slightly improved by using a more involved algorithm, e.g. a variation of the Bron-Kerbosch algorithm [58], however, the time needed for the facet computation outweighs the time needed for finding the stable sets by far, which is why we deem this optimization not necessary.

of constraints. Lastly, if the graph structure matches, we verify whether the coefficients of a and b also match.

To make the approach outlined in the previous paragraph more concrete, we explain the case of odd hole constraints in more depth: Checking the graph structure in this case consists of checking that $|S|$ is odd, that every vertex has degree 2, and that the graph is connected (this last step can technically be omitted, since non-connected graphs cannot induce facets). To check the coefficients we just need to check $a_v = 1$ for $v \in S$ and $b = (|S| - 1)/2$.

The other types of constraints can be handled in a similar manner, although some require a little more consideration due to the role of special vertices like the hub of an odd wheel. For the sake of brevity, we omit a detailed explanation of the algorithms that we devised for the other types.

Our complete computational pipeline can thus be summarized in the following steps:

- (i) Sample G from $G(n, m; p)$,
- (ii) compute all stable sets in G ,
- (iii) compute all facet-defining inequalities of $\text{STAB}(G)$,
- (iv) classify all facet-defining inequalities.

6.2 Computational results

We sampled 50 graphs for each combination of $n, m \geq 1$ with the resulting graph having at most 20 vertices, i.e. $n + m \leq 18$, and for each value $p \in \{0.25, 0.5, 0.75\}$ from $G(n, m; p)$. In Figure 6.1, the mean percentage of facets that did not fall into the classes (i) to (v) of Theorem 5.15 or the class of odd-subdivided odd wheel constraints for each of the parameter combinations is displayed. The underlying numerical data, including the standard deviations of the values, can be found in the appendix in Tables A.1, A.2, and A.3.

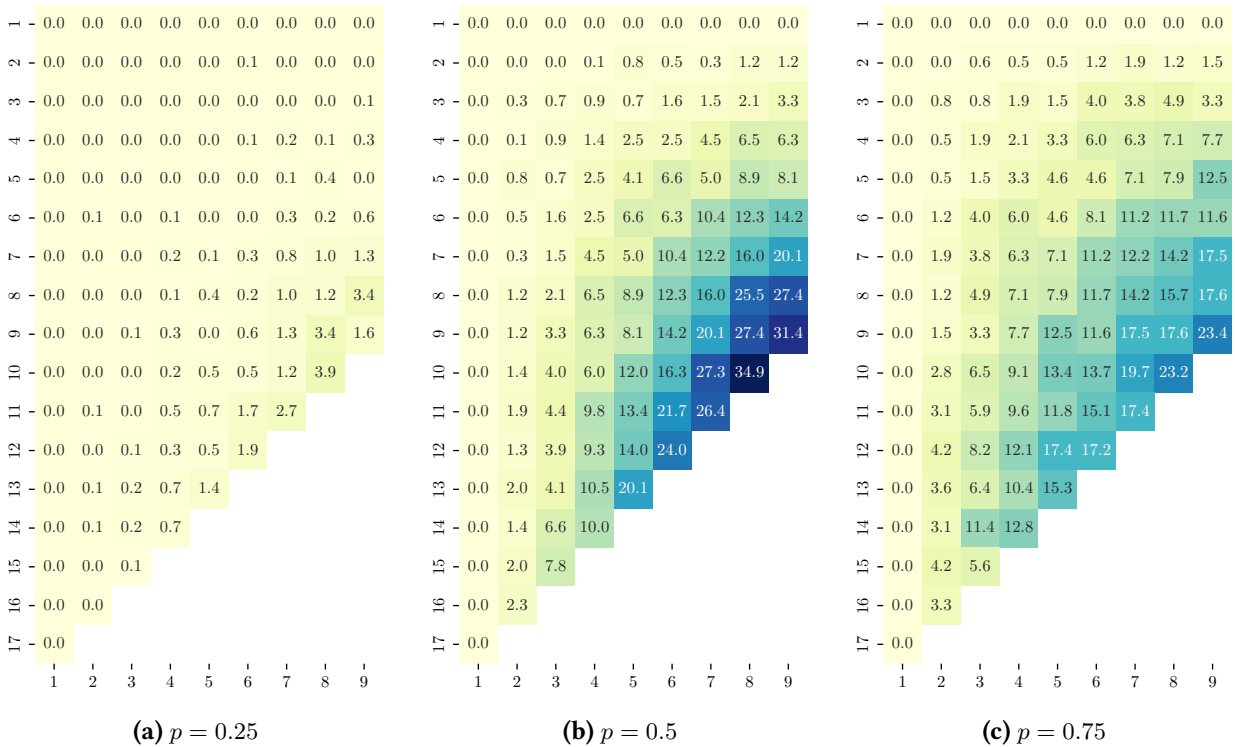


Figure 6.1 Mean percentage of unclassified facets for $G(n, m; p)$ based on 50 samples for different values of p and all values $n, m \geq 1$ such that the resulting graph has at most 20 vertices. Rows are indexed by n , columns by m .

Observe that for relatively sparse graphs, i. e. those with $p = 0.25$, almost all facets belong to the known types, even for the largest graphs that we considered. Similarly, for relatively dense graphs, i. e. those with $p = 0.75$ a relatively large share of facets belong to the known types. This trend is further supported by other, smaller-scale experiments that we conducted: Very sparse and very dense graphs have almost all facets in the known classes. Moreover, at the extremes $p = 0$ and $p = 1$, all facets belong to the known classes, since in this case $G(n, m; p)$ is (almost surely) the complement of a quasi-line graph and hence Theorem 5.15 is applicable. However, as our experiment also indicates, on average, there seem to be more unclassified constraints for a given value p relatively close to 1 than for the corresponding value $1 - p$ for the same values of (n, m) . Therefore, the most interesting, and in some sense worst case in our experiment is $p = 0.5$.

Focusing on this case $p = 0.5$, we further see that among different values (n, m) with the same sum $n + m$, the highest average percentage of unclassified facets occurs when n and m are equal or close to equal. Therefore, we will focus on the case $n = m$ in the remainder of our analysis. In Figure 6.2 the average composition of the facets of $G(n, n; 0.5)$ for $n \in \{1, \dots, 9\}$ is displayed. In each column, the mean share of the total number of facets that fall into each of the aforementioned categories for the given n is displayed. The underlying numerical data, including the standard deviations of the values, can be found in the appendix in Table A.4.

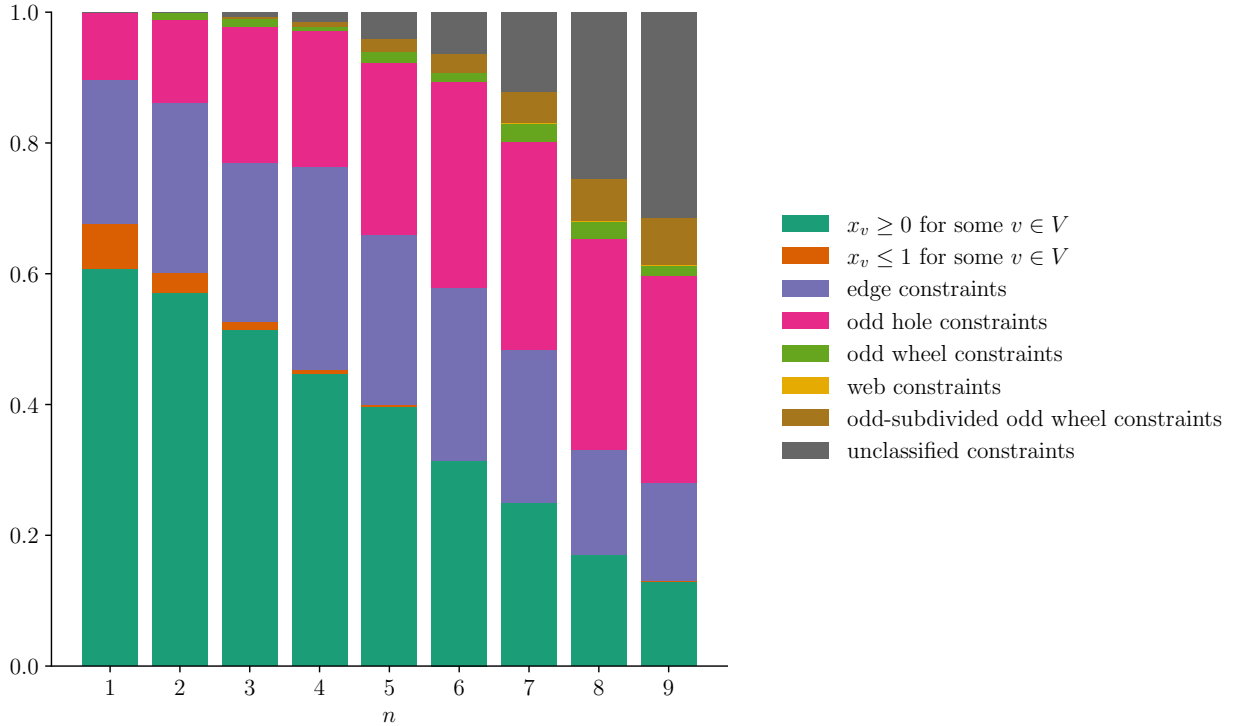


Figure 6.2 Average composition of facets for $G(n, n; 0.5)$ based on 50 samples for $n \in \{1, \dots, 9\}$. Each section represents the mean share of total facets that fall into the indicated category for the respective n .

The first thing we want to highlight is how the share of each class of facets changes as n changes. Note that the non-negativity constraints start with the largest share for small n , which then gradually decreases as n increases. This is because the number of such facets is linear in n ($2n + 2$ to be precise, one for each vertex in $G(n, n; p)$), while the average total number of facets seems to be exponential, as is displayed in Figure 6.3 (the numerical data for which can be found in Table A.5). Note further that the share of constraints of the form $x_v \leq 1$ goes to 0 as n increases, which is to be expected, since such a constraint is facet-defining if and only if v is an isolated vertex, i. e. has degree 0, the probability of which goes to 0 as n goes to infinity. For edge constraints, no clear trend is visible. They maintain a relatively constant share across all examined values of n . In contrast, odd hole constraints seem to take up a slightly larger share as n increases. The same is true for odd wheel constraints, especially when grouped with the odd-subdivided

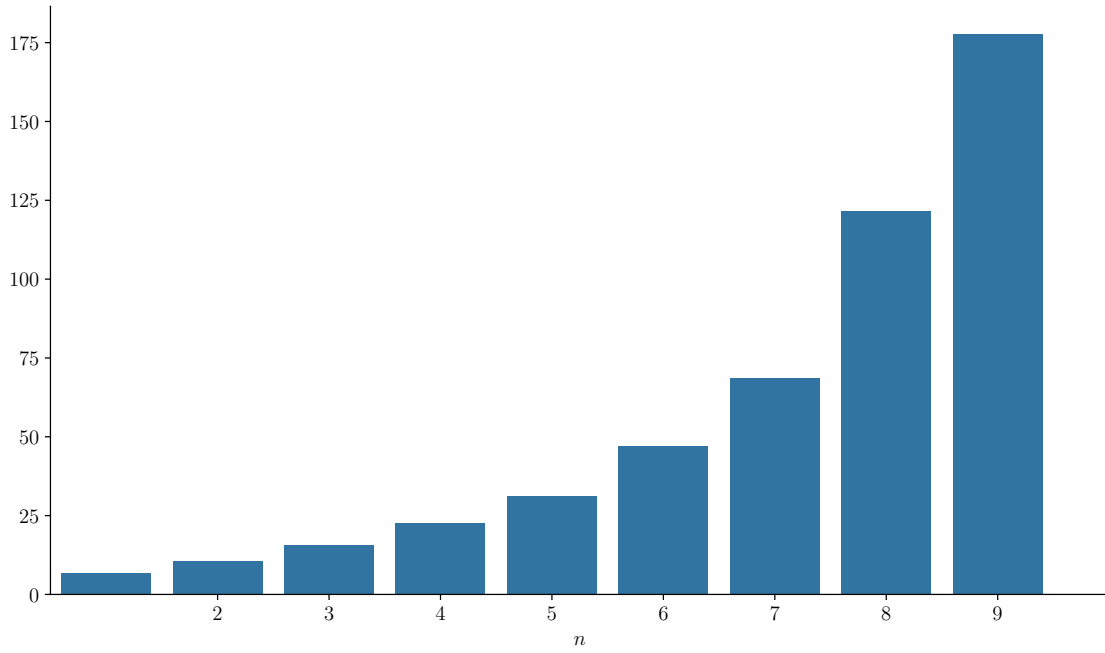


Figure 6.3 Average total number of facets for $G(n, n; 0.5)$ based on 50 samples for $n \in \{1, \dots, 9\}$.

odd wheel constraints. Lastly, web constraints seem to appear very rarely, and they appear only for $n \geq 8$ as the smallest web that is relevant is $W(8, 3)$.

Moreover, note that the share of unclassified constraints, i.e. the portion of constraints that do not belong to any of the categories mentioned before, increases as n increases. Unfortunately, this increase is quite significant, and, based on the available data, it appears to be superlinear or even exponential. However, the available data does not suffice to draw any definitive conclusions about the behavior as n goes to infinity. That being said, the available data strongly suggests that constraints of types (i) to (v) miss a significant share of facets, which becomes larger as the size of the graphs increases.

6.3 Challenges of the computational approach

One natural approach based on the results of the previous section would be to inspect the facets that were left unclassified manually and to try to attribute them to another group of constraints, either known from the literature or a novel family of constraints. As mentioned earlier, we were unable to do this for any further constraints. However, it is quite likely that within the vast amount of literature on stable set polytopes, there are other known families of constraints relevant to our setting that we did not find. It is consequently quite likely that one can increase the number of constraints belonging to explicitly known families beyond what was presented in the previous section in this way. Even so, in this section, we aim to present some evidence that this approach may inherently not suffice to explain all facets of general graphs with odd cycle transversal number 2.

The first point we want to make is based on the observation that inequalities corresponding to non-induced subgraphs can be facet-defining for the stable set polytope of the full graph. This can be seen in the examples given in Figure 6.4. To correctly classify these constraints into the known families, we would need to answer the question whether, after deletion of some edges of the original graph, it is isomorphic to some graph in the family of graphs underlying a family of constraints. Answering this question is closely related to solving a subgraph isomorphism problem where the subgraph has to be a spanning subgraph, i.e. use all vertices of the original graph, a problem which is known to be NP-hard [3]. More concretely, we have the following observation.

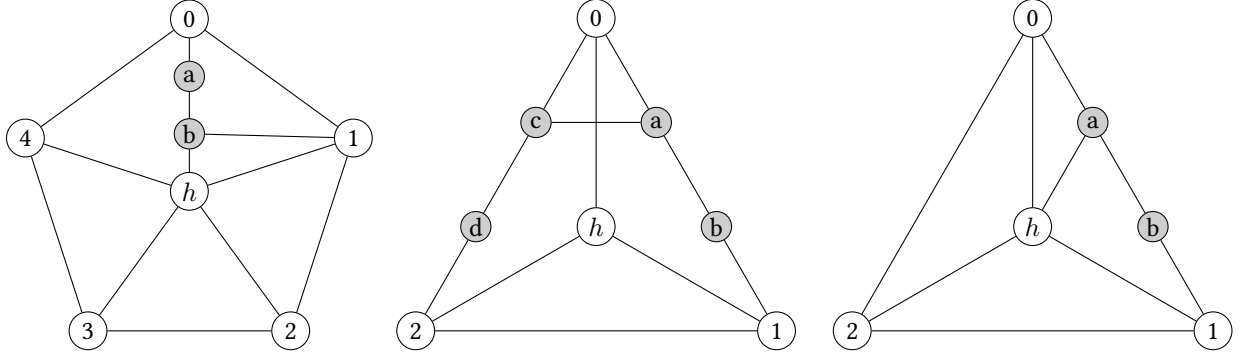


Figure 6.4 Three examples of graphs which have facet defining odd-subdivided odd wheel constraints induced by a non-induced subgraph.

Proposition 6.1. *Given a graph $G = (V, E)$, it is NP-hard to decide whether there is $E' \subseteq E$ such that $H = (V, E')$ is an odd wheel.*

Proof. Consider the Hamiltonian path problem with fixed endpoints, i. e. given a graph $\tilde{G} = (\tilde{V}, \tilde{E})$ and vertices $s, t \in \tilde{V}$ decide whether there exists an s - t -path in $\tilde{G}$ that uses all vertices. This problem is known to be NP-hard [3]. We reduce this problem to the given one. To this end, we introduce additional vertices $S = \{h, v_1, \dots, v_k\}$, where $k = 3$ if $|\tilde{V}|$ is even and $k = 2$ if $|\tilde{V}|$ is odd, and additional edges $T = \{\{h, w\} : w \in \tilde{V} \cup \{v_1, \dots, v_k\}\} \cup \{\{v_i, v_{i+1} : 1 \leq i < k\}\} \cup \{\{s, v_1\}, \{t, v_k\}\}$. The idea is to have h as the hub and to ensure this is the only candidate for a hub, since it is the only vertex connected to all other vertices. Moreover, a Hamiltonian s - t -path can be extended to an odd cycle through the edges $\{\{v_i, v_{i+1} : 1 \leq i < k\}\} \cup \{\{s, v_1\}, \{t, v_k\}\}$.

Formally, we define $G = (V, E)$ with $V = \tilde{V} \cup S$ and $E = \tilde{E} \cup T$. If $\tilde{G}$ has a Hamiltonian s - t -path P with edge set $E(P)$, choosing $E' = E(P) \cup T$ yields an odd wheel $H = (V, E')$ with hub h . Conversely, let $E' \subseteq E$ be a set of edges such that $H = (V, E')$ is an odd wheel. Denote the hub of this wheel by h' . Since the hub of an odd wheel is connected to all other vertices, we need to have $h = h'$. Deleting h from H must yield an odd cycle that uses all vertices in $V \setminus \{h\}$. Now, since $v_1, \dots, v_k$ have degree 2 in the graph that results from H by deleting h , we see that any such cycle must contain the path with vertices $s, v_1, \dots, v_k, t$ in this order and the remainder of the odd cycle must be a Hamiltonian s - t -path in G . $\square$

This result indicates that the classification of non-induced odd wheel constraints might be hard. There are, however, two important shortcomings of the result: First, it does not show that the problem remains NP-hard when restricting G to have odd cycle transversal number 2, and second, it does not show that the problem remains NP-hard when requiring instead that H is an odd-subdivided odd wheel. Nevertheless, it provides some evidence that dealing with non-induced subgraphs poses a challenge.

The second point we want to make is to highlight the challenge resulting from Proposition 5.4. This proposition allows us to generate much larger classes of inequalities from known classes. However, algorithmically detecting whether a given constraint arose from a known constraint by a sequence of identifications of vertices is a challenging problem.

Both of these observations point towards the fact that among the unclassified facets are some, potentially many, that relate to known constraints in a simple way on a theoretical level, but detecting this relation computationally is hard. We are therefore in a situation in which we would like to consider some unclassified constraints as belonging to a known group, but are unable to do so computationally. It is unclear to us how one would overcome this issue. We therefore believe that the more promising path towards a complete description of the stable set polytope of graphs with odd cycle transversal number 2 is the path based on further theoretical work.

7 Outlook

We have shown that the Balas-Ceria-Cornuéjols hierarchy, and consequently also the Lovász-Schrijver, Sherali-Adams, and Lasserre hierarchies, applied to the fractional stable set polytope achieve an integrality gap of $1 + \varepsilon$ after a number of rounds that depends only on ε and the odd cycle packing number. Moreover, we have established that these hierarchies reach the integer hull exactly after a number of rounds depending only on the odd cycle transversal number. This naturally raises the question of whether the number of rounds required for the hierarchies to reach the integer hull exactly can also be bounded in terms of the odd cycle packing number. It is also open whether the hierarchies differ in this respect. Moreover, as a related question, it remains unclear whether the CG closure admits comparable results, both regarding achieving an arbitrarily small integrality gap and reaching the integer hull exactly.

In addition, we have seen a complete linear description of the stable set polytope of graphs with odd cycle transversal number 2 and 3 that are complements of quasi-line graphs, and we have examined the shortcomings of this description for the stable set polytope of general graphs with odd cycle transversal number 2. It would be interesting to obtain a complete linear description also in the case of general graphs with odd cycle transversal number 2.

Finally, we have shown that in the case of complements of quasi-line graphs, the CG rank can be bounded by the odd cycle transversal number if the latter is at most three. A direction for future research would be to examine whether this also holds for larger odd cycle transversal number and whether it extends to the general case, beyond complements of quasi-line graphs.

A Appendix: Computational Data

Table A.1 Mean percentage $\pm$ standard deviation of unclassified facets for $G(n, m; 0.25)$ based on 50 samples. The rows are labeled by n , the columns by m .

	1	2	3	4	5	6	7	8	9
1	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
2	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
3	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.8
4	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.7	0.2 $\pm$ 1.1	0.1 $\pm$ 0.5	0.3 $\pm$ 1.2
5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.4	0.4 $\pm$ 1.6	0.0 $\pm$ 0.3
6	0.0 $\pm$ 0.0	0.1 $\pm$ 0.6	0.0 $\pm$ 0.0	0.1 $\pm$ 0.7	0.0 $\pm$ 0.0	0.0 $\pm$ 0.3	0.3 $\pm$ 0.9	0.2 $\pm$ 0.8	0.6 $\pm$ 1.9
7	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.2 $\pm$ 1.1	0.1 $\pm$ 0.4	0.3 $\pm$ 0.9	0.8 $\pm$ 2.1	1.0 $\pm$ 3.0	1.3 $\pm$ 3.0
8	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.5	0.4 $\pm$ 1.6	0.2 $\pm$ 0.8	1.0 $\pm$ 3.0	1.2 $\pm$ 3.1	3.4 $\pm$ 6.9
9	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.8	0.3 $\pm$ 1.2	0.0 $\pm$ 0.3	0.6 $\pm$ 1.9	1.3 $\pm$ 3.0	3.4 $\pm$ 6.9	1.6 $\pm$ 3.5
10	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.2 $\pm$ 0.9	0.5 $\pm$ 1.7	0.5 $\pm$ 1.4	1.2 $\pm$ 2.8	3.9 $\pm$ 6.8	–
11	0.0 $\pm$ 0.0	0.1 $\pm$ 0.9	0.0 $\pm$ 0.0	0.5 $\pm$ 1.5	0.7 $\pm$ 1.9	1.7 $\pm$ 4.1	2.7 $\pm$ 6.0	–	–
12	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.6	0.3 $\pm$ 1.0	0.5 $\pm$ 2.2	1.9 $\pm$ 3.4	–	–	–
13	0.0 $\pm$ 0.0	0.1 $\pm$ 0.4	0.2 $\pm$ 0.8	0.7 $\pm$ 1.8	1.4 $\pm$ 2.8	–	–	–	–
14	0.0 $\pm$ 0.0	0.1 $\pm$ 0.4	0.2 $\pm$ 0.7	0.7 $\pm$ 2.0	–	–	–	–	–
15	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.5	–	–	–	–	–	–
16	0.0 $\pm$ 0.0	0.0 $\pm$ 0.3	–	–	–	–	–	–	–
17	0.0 $\pm$ 0.0	–	–	–	–	–	–	–	–

Table A.2 Mean percentage $\pm$ standard deviation of unclassified facets for $G(n, m; 0.5)$ based on 50 samples. The rows are labeled by n , the columns by m .

	1	2	3	4	5	6	7	8	9
1	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
2	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.1 $\pm$ 0.8	0.8 $\pm$ 2.1	0.5 $\pm$ 1.6	0.3 $\pm$ 1.3	1.2 $\pm$ 2.3	1.2 $\pm$ 2.7
3	0.0 $\pm$ 0.0	0.3 $\pm$ 1.3	0.7 $\pm$ 2.0	0.9 $\pm$ 2.1	0.7 $\pm$ 2.2	1.6 $\pm$ 2.9	1.5 $\pm$ 3.3	2.1 $\pm$ 3.5	3.3 $\pm$ 4.3
4	0.0 $\pm$ 0.0	0.1 $\pm$ 0.8	0.9 $\pm$ 2.1	1.4 $\pm$ 2.5	2.5 $\pm$ 4.0	2.5 $\pm$ 3.4	4.5 $\pm$ 5.7	6.5 $\pm$ 7.1	6.3 $\pm$ 5.8
5	0.0 $\pm$ 0.0	0.8 $\pm$ 2.1	0.7 $\pm$ 2.2	2.5 $\pm$ 4.0	4.1 $\pm$ 5.6	6.6 $\pm$ 6.4	5.0 $\pm$ 6.4	8.9 $\pm$ 8.2	8.1 $\pm$ 7.0
6	0.0 $\pm$ 0.0	0.5 $\pm$ 1.6	1.6 $\pm$ 2.9	2.5 $\pm$ 3.4	6.6 $\pm$ 6.4	6.3 $\pm$ 6.4	10.4 $\pm$ 8.2	12.3 $\pm$ 11.1	14.2 $\pm$ 9.5
7	0.0 $\pm$ 0.0	0.3 $\pm$ 1.3	1.5 $\pm$ 3.3	4.5 $\pm$ 5.7	5.0 $\pm$ 6.4	10.4 $\pm$ 8.2	12.2 $\pm$ 9.6	16.0 $\pm$ 11.3	20.1 $\pm$ 13.0
8	0.0 $\pm$ 0.0	1.2 $\pm$ 2.3	2.1 $\pm$ 3.5	6.5 $\pm$ 7.1	8.9 $\pm$ 8.2	12.3 $\pm$ 11.1	16.0 $\pm$ 11.3	25.5 $\pm$ 13.7	27.4 $\pm$ 12.0
9	0.0 $\pm$ 0.0	1.2 $\pm$ 2.7	3.3 $\pm$ 4.3	6.3 $\pm$ 5.8	8.1 $\pm$ 7.0	14.2 $\pm$ 9.5	20.1 $\pm$ 13.0	27.4 $\pm$ 12.0	31.4 $\pm$ 13.8
10	0.0 $\pm$ 0.0	1.4 $\pm$ 5.2	4.0 $\pm$ 5.6	6.0 $\pm$ 7.8	12.0 $\pm$ 9.0	16.3 $\pm$ 13.5	27.3 $\pm$ 13.0	34.9 $\pm$ 15.3	–
11	0.0 $\pm$ 0.0	1.9 $\pm$ 3.6	4.4 $\pm$ 7.2	9.8 $\pm$ 9.4	13.4 $\pm$ 10.8	21.7 $\pm$ 13.6	26.4 $\pm$ 13.0	–	–
12	0.0 $\pm$ 0.0	1.3 $\pm$ 2.5	3.9 $\pm$ 5.5	9.3 $\pm$ 8.7	14.0 $\pm$ 11.9	24.0 $\pm$ 13.5	–	–	–
13	0.0 $\pm$ 0.0	2.0 $\pm$ 3.4	4.1 $\pm$ 5.8	10.5 $\pm$ 8.3	20.1 $\pm$ 10.9	–	–	–	–
14	0.0 $\pm$ 0.0	1.4 $\pm$ 2.7	6.6 $\pm$ 8.9	10.0 $\pm$ 8.0	–	–	–	–	–
15	0.0 $\pm$ 0.0	2.0 $\pm$ 3.5	7.8 $\pm$ 9.1	–	–	–	–	–	–
16	0.0 $\pm$ 0.0	2.3 $\pm$ 4.7	–	–	–	–	–	–	–
17	0.0 $\pm$ 0.0	–	–	–	–	–	–	–	–

Table A.3 Mean percentage $\pm$ standard deviation of unclassified facets for $G(n, m; 0.75)$ based on 50 samples. The rows are labeled by n , the columns by m .

	1	2	3	4	5	6	7	8	9
1	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
2	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.6 $\pm$ 2.0	0.5 $\pm$ 1.9	0.5 $\pm$ 2.0	1.2 $\pm$ 2.7	1.9 $\pm$ 3.4	1.2 $\pm$ 2.4	1.5 $\pm$ 3.6
3	0.0 $\pm$ 0.0	0.8 $\pm$ 2.7	0.8 $\pm$ 2.2	1.9 $\pm$ 4.3	1.5 $\pm$ 2.6	4.0 $\pm$ 6.1	3.8 $\pm$ 4.8	4.9 $\pm$ 7.1	3.3 $\pm$ 5.1
4	0.0 $\pm$ 0.0	0.5 $\pm$ 1.9	1.9 $\pm$ 4.3	2.1 $\pm$ 3.7	3.3 $\pm$ 4.0	6.0 $\pm$ 7.2	6.3 $\pm$ 8.1	7.1 $\pm$ 7.5	7.7 $\pm$ 7.5
5	0.0 $\pm$ 0.0	0.5 $\pm$ 2.0	1.5 $\pm$ 2.6	3.3 $\pm$ 4.0	4.6 $\pm$ 5.4	4.6 $\pm$ 4.3	7.1 $\pm$ 9.2	7.9 $\pm$ 8.6	12.5 $\pm$ 10.2
6	0.0 $\pm$ 0.0	1.2 $\pm$ 2.7	4.0 $\pm$ 6.1	6.0 $\pm$ 7.2	4.6 $\pm$ 4.3	8.1 $\pm$ 7.6	11.2 $\pm$ 10.9	11.7 $\pm$ 9.3	11.6 $\pm$ 9.4
7	0.0 $\pm$ 0.0	1.9 $\pm$ 3.4	3.8 $\pm$ 4.8	6.3 $\pm$ 8.1	7.1 $\pm$ 9.2	11.2 $\pm$ 10.9	12.2 $\pm$ 9.5	14.2 $\pm$ 10.9	17.5 $\pm$ 11.7
8	0.0 $\pm$ 0.0	1.2 $\pm$ 2.4	4.9 $\pm$ 7.1	7.1 $\pm$ 7.5	7.9 $\pm$ 8.6	11.7 $\pm$ 9.3	14.2 $\pm$ 10.9	15.7 $\pm$ 11.6	17.6 $\pm$ 11.6
9	0.0 $\pm$ 0.0	1.5 $\pm$ 3.6	3.3 $\pm$ 5.1	7.7 $\pm$ 7.5	12.5 $\pm$ 10.2	11.6 $\pm$ 9.4	17.5 $\pm$ 11.7	17.6 $\pm$ 11.6	23.4 $\pm$ 13.2
10	0.0 $\pm$ 0.0	2.8 $\pm$ 4.4	6.5 $\pm$ 7.2	9.1 $\pm$ 10.0	13.4 $\pm$ 9.3	13.7 $\pm$ 11.6	19.7 $\pm$ 12.3	23.2 $\pm$ 14.7	–
11	0.0 $\pm$ 0.0	3.1 $\pm$ 5.3	5.9 $\pm$ 5.9	9.6 $\pm$ 9.8	11.8 $\pm$ 9.6	15.1 $\pm$ 11.7	17.4 $\pm$ 10.6	–	–
12	0.0 $\pm$ 0.0	4.2 $\pm$ 7.6	8.2 $\pm$ 7.4	12.1 $\pm$ 10.6	17.4 $\pm$ 11.1	17.2 $\pm$ 12.6	–	–	–
13	0.0 $\pm$ 0.0	3.6 $\pm$ 5.5	6.4 $\pm$ 7.8	10.4 $\pm$ 9.7	15.3 $\pm$ 13.1	–	–	–	–
14	0.0 $\pm$ 0.0	3.1 $\pm$ 4.9	11.4 $\pm$ 10.1	12.8 $\pm$ 11.0	–	–	–	–	–
15	0.0 $\pm$ 0.0	4.2 $\pm$ 8.2	5.6 $\pm$ 7.8	–	–	–	–	–	–
16	0.0 $\pm$ 0.0	3.3 $\pm$ 6.2	–	–	–	–	–	–	–
17	0.0 $\pm$ 0.0	–	–	–	–	–	–	–	–

Table A.4 Average composition of facets for $G(n, n; 0.5)$ based on 50 samples for $n \in \{1, \dots, 9\}$. Each cell reports the mean percentage $\pm$ the standard deviation of the share of total facets that fall into the indicated category for the respective n . The rows are labeled by n .

	$x_v \geq 0$	$x_v \leq 1$	edge	odd hole	odd wheel	web	subd. odd wheel	unclassified
1	60.9 $\pm$ 5.2	6.8 $\pm$ 10.3	22.0 $\pm$ 16.0	10.3 $\pm$ 13.8	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
2	57.1 $\pm$ 4.9	3.0 $\pm$ 6.2	25.9 $\pm$ 15.2	12.8 $\pm$ 11.9	1.1 $\pm$ 4.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
3	51.5 $\pm$ 6.3	1.2 $\pm$ 3.3	24.3 $\pm$ 13.4	20.7 $\pm$ 11.7	1.3 $\pm$ 2.9	0.0 $\pm$ 0.0	0.2 $\pm$ 1.0	0.7 $\pm$ 2.0
4	44.7 $\pm$ 5.5	0.6 $\pm$ 1.7	31.1 $\pm$ 13.6	20.7 $\pm$ 11.2	0.7 $\pm$ 2.9	0.0 $\pm$ 0.0	0.8 $\pm$ 2.5	1.4 $\pm$ 2.5
5	39.7 $\pm$ 6.8	0.2 $\pm$ 0.7	26.0 $\pm$ 9.3	26.4 $\pm$ 7.4	1.7 $\pm$ 3.3	0.0 $\pm$ 0.0	1.9 $\pm$ 2.9	4.1 $\pm$ 5.6
6	31.4 $\pm$ 6.7	0.0 $\pm$ 0.3	26.4 $\pm$ 9.0	31.5 $\pm$ 9.5	1.4 $\pm$ 2.7	0.0 $\pm$ 0.0	3.0 $\pm$ 3.6	6.3 $\pm$ 6.4
7	25.0 $\pm$ 6.4	0.0 $\pm$ 0.2	23.4 $\pm$ 10.4	31.8 $\pm$ 5.9	2.8 $\pm$ 6.2	0.0 $\pm$ 0.0	4.8 $\pm$ 4.8	12.2 $\pm$ 9.6
8	17.0 $\pm$ 5.8	0.0 $\pm$ 0.0	16.1 $\pm$ 6.3	32.2 $\pm$ 8.1	2.7 $\pm$ 4.9	0.1 $\pm$ 0.3	6.4 $\pm$ 4.2	25.5 $\pm$ 13.7
9	13.0 $\pm$ 4.5	0.0 $\pm$ 0.0	15.0 $\pm$ 6.2	31.8 $\pm$ 8.5	1.6 $\pm$ 3.0	0.1 $\pm$ 0.3	7.3 $\pm$ 4.9	31.4 $\pm$ 13.8

Table A.5 Mean $\pm$ standard deviation of the total number of facets of $G(n, n; 0.5)$ based on 50 samples for $n \in \{1, \dots, 9\}$. The rows are labeled by n .

	Mean $\pm$ standard deviation
1	6.62 $\pm$ 0.57
2	10.58 $\pm$ 0.93
3	15.78 $\pm$ 1.99
4	22.70 $\pm$ 2.81
5	31.16 $\pm$ 5.77
6	46.96 $\pm$ 11.44
7	68.54 $\pm$ 19.24
8	121.48 $\pm$ 51.48
9	177.64 $\pm$ 75.81

Bibliography

- [1] A. Bock, Y. Faenza, C. Moldenhauer, and A. J. Ruiz-Vargas. “Solving the Stable Set Problem in Terms of the Odd Cycle Packing Number”. In: *34th International Conference on Foundation of Software Technology and Theoretical Computer Science (FSTTCS 2014)*. Vol. 29. Leibniz International Proceedings in Informatics (LIPIcs). Wadern, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2014, pp. 187–198.
- [2] F. B. Shepherd. “Applying Lehman’s theorems to packing problems”. In: *Mathematical Programming* 71 (1995), pp. 353–367.
- [3] M. R. Garey and D. S. Johnson. *Computers and Intractability; A Guide to the Theory of NP-Completeness*. W. H. Freeman & Co., 1990.
- [4] S. Fiorini, G. Joret, S. Weltge, and Y. Yuditsky. “Integer programs with bounded subdeterminants and two nonzeros per row”. In: *2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS)*. 2022, pp. 13–24.
- [5] E. Györi, A. V. Kostochka, and T. Łuczak. “Graphs without short odd cycles are nearly bipartite”. In: *Discrete Mathematics* 163.1 (1997), pp. 279–284.
- [6] B. Reed. “Mangoes and Blueberries”. In: *Combinatorica* 19.2 (Feb. 1999), pp. 267–296.
- [7] J. W. Grossman, D. M. Kulkarni, and I. E. Schochetman. “On the minors of an incidence matrix and its Smith normal form”. In: *Linear Algebra and its Applications* 218 (1995), pp. 213–224.
- [8] E. Balas, S. Ceria, and G. Cornuéjols. “A lift-and-project cutting plane algorithm for mixed 0–1 programs”. In: *Mathematical Programming* 58.1 (Jan. 1993), pp. 295–324.
- [9] M. Laurent. “A Comparison of the Sherali-Adams, Lovász-Schrijver, and Lasserre Relaxations for 0–1 Programming”. In: *Mathematics of Operations Research* 28.3 (2003), pp. 470–496.
- [10] M. Conforti, G. Cornuéjols, and G. Zambelli. *Integer Programming*. Vol. 271. Graduate Texts in Mathematics. Springer, 2014.
- [11] M. Conforti, G. Cornuéjols, and G. Zambelli. “Extended formulations in combinatorial optimization”. In: *4OR* 8.1 (Mar. 2010), pp. 1–48.
- [12] V. Chvátal. “On certain polytopes associated with graphs”. In: *Journal of Combinatorial Theory, Series B* 18.2 (1975), pp. 138–154.
- [13] R. E. Gomory. “Outline of an algorithm for integer solutions to linear programs”. In: *Bulletin of the American Mathematical Society* 64.5 (Sept. 1958), pp. 275–278.
- [14] R. E. Gomory. “An algorithm for integer solutions to linear programs”. In: *Recent Advances in Mathematical Programming*. McGraw-Hill, 1963, pp. 269–302.
- [15] F. Eisenbrand. “Note – On the membership problem for the elementary closure of a polyhedron”. In: *Combinatorica* 19.2 (Feb. 1999), pp. 297–300.
- [16] M. Grötschel, L. Lovász, and A. Schrijver. “The ellipsoid method and its consequences in combinatorial optimization”. In: *Combinatorica* 1.2 (June 1981), pp. 169–197.
- [17] V. Chvátal. “Edmonds polytopes and a hierarchy of combinatorial problems”. In: *Discrete Mathematics* 4.4 (1973), pp. 305–337.
- [18] A. Schrijver. “On Cutting Planes”. In: *Combinatorics* 79. Vol. 9. Annals of Discrete Mathematics. Elsevier, 1980, pp. 291–296.

- [19] F. Eisenbrand and A. S. Schulz. “Bounds on the Chvátal rank of polytopes in the 0/1-cube”. In: *Combinatorica* 23.2 (2003), pp. 245–261.
- [20] V. Chvátal, W. Cook, and M. Hartmann. “On cutting-plane proofs in combinatorial optimization”. In: *Linear Algebra and its Applications* 114–115 (1989). Special Issue Dedicated to Alan J. Hoffman, pp. 455–499.
- [21] T. Rothvoß and L. Sanità. “0/1 Polytopes with Quadratic Chvátal Rank”. In: *Operations Research* 65.1 (2017), pp. 212–220.
- [22] A. Kurpisz, S. Leppänen, and M. Mastrolilli. “On the Hardest Problem Formulations for the 0/1 Lasserre Hierarchy”. In: *Automata, Languages, and Programming*. Berlin, Heidelberg: Springer-Verlag, 2015, pp. 872–885.
- [23] J. Komlós. “Covering odd cycles”. In: *Combinatorica* 17.3 (1997), pp. 393–400.
- [24] S. Har-Peled and S. Rahul. “Two (Known) Results About Graphs with No Short Odd Cycles”. In: *arXiv* (2018).
- [25] M. W. Padberg. “On the facial structure of set packing polyhedra”. In: *Mathematical Programming* 5 (1973), pp. 199–215.
- [26] M. Grötschel, L. Lovász, and A. Schrijver. “Chapter 9: Stable sets in graphs”. In: *Geometric Algorithms and Combinatorial Optimization*. 2nd ed. Algorithms and Combinatorics. Berlin, Heidelberg: Springer-Verlag, 1993, pp. 272–299.
- [27] M. Chudnovsky, N. Robertson, P. Seymour, and R. Thomas. “The Strong Perfect Graph Theorem”. In: *Annals of Mathematics* 164.1 (2006), pp. 51–229.
- [28] L. Lovász. “Normal hypergraphs and the perfect graph conjecture”. In: *Discrete Mathematics* 2.3 (1972), pp. 253–267.
- [29] S. Hougardy. “Classes of perfect graphs”. In: *Discrete Mathematics* 306.19 (2006), pp. 2529–2571.
- [30] L. Lovász. “On the Shannon capacity of a graph”. In: *IEEE Transactions on Information Theory* 25.1 (1979), pp. 1–7.
- [31] J. Fonlupt and J. Uhry. “Transformations which Preserve Perfectness and H-Perfectness of Graphs”. In: *Bonn Workshop on Combinatorial Optimization*. Vol. 66. North-Holland Mathematics Studies. North-Holland, 1982, pp. 83–95.
- [32] M. Campêlo and G. Cornuéjols. “Stable sets, corner polyhedra and the Chvátal closure”. In: *Operations Research Letters* 37.6 (2009), pp. 375–378.
- [33] E. Holm, L. M. Torres, and A. K. Wagler. “On the Chvátal rank of linear relaxations of the stable set polytope”. In: *International Transactions in Operational Research* 17.6 (2010), pp. 827–849.
- [34] N. Sbihi and J. Uhry. “A class of h-perfect graphs”. In: *Discrete Mathematics* 51.2 (1984), pp. 191–205.
- [35] M. Grötschel, L. Lovász, and A. Schrijver. “Relaxations of vertex packing”. In: *Journal of Combinatorial Theory, Series B* 40.3 (1986), pp. 330–343.
- [36] L. Lovász and A. Schrijver. “Cones of Matrices and Set-Functions and 0–1 Optimization”. In: *SIAM Journal on Optimization* 1.2 (1991), pp. 166–190.
- [37] G. L. Nemhauser and L. E. Trotter. “Properties of vertex packing and independence system polyhedra”. In: *Mathematical Programming* 6 (1974), pp. 48–61.
- [38] M. W. Padberg. “Technical Note—A Note on Zero-One Programming”. In: *Operations Research* 23.4 (1975), pp. 833–837.
- [39] L. Trotter. “A class of facet producing graphs for vertex packing polyhedra”. In: *Discrete Mathematics* 12.4 (1975), pp. 373–388.
- [40] G. Oriolo. “Clique family inequalities for the stable set polytope of quasi-line graphs”. In: *Discrete Applied Mathematics* 132.1 (2003). Stability in Graphs and Related Topics, pp. 185–201.

- [41] A. Pecher and A. K. Wagler. “A note on the Chvátal rank of clique family inequalities”. In: *RAIRO - Operations Research* 41 (2007), pp. 289–294.
- [42] A. N. Letchford and P. Ventura. “Strengthened clique-family inequalities for the stable set polytope”. In: *Operations Research Letters* 49.4 (2021), pp. 586–589.
- [43] R. Borndörfer and R. Weismantel. “Set packing relaxations of some integer programs”. In: *Mathematical Programming* 88 (2000), pp. 425–450.
- [44] E. Cheng and W. H. Cunningham. “Wheel inequalities for stable set polytopes”. In: *Mathematical Programming* 77 (1997), pp. 389–421.
- [45] E. Cheng and S. de Vries. “Antiweb-wheel inequalities and their separation problems over the stable set polytopes”. In: *Mathematical Programming* 92 (2002), pp. 153–175.
- [46] F. Harary and R. Z. Norman. “Some properties of line digraphs”. In: *Rendiconti del Circolo Matematico di Palermo* 9 (1960), pp. 161–168.
- [47] J. Edmonds. “Maximum matching and a polyhedron with 0,1-vertices”. In: *Journal of Research of the National Bureau of Standards Section B Mathematics and Mathematical Physics* (1965), p. 125.
- [48] A. Ben Rebea. “Étude des stables dans les graphes quasi-adjoints”. PhD Thesis. Université de Grenoble, 1981.
- [49] F. Eisenbrand, G. Oriolo, G. Stauffer, and P. Ventura. “The stable set polytope of quasi-line graphs”. In: *Combinatorica* 28 (2008), pp. 45–67.
- [50] R. Faudree, E. Flandrin, and Z. Ryjáček. “Claw-free graphs — A survey”. In: *Discrete Mathematics* 164.1 (1997). The second Krakow conference of graph theory, pp. 87–147.
- [51] R. Giles and L. Trotter. “On stable set polyhedra for $K_{1,3}$ -free graphs”. In: *Journal of Combinatorial Theory, Series B* 31.3 (1981), pp. 313–326.
- [52] A. Lehman. “On the width—length inequality”. In: *Mathematical Programming* 17 (1979), pp. 403–417.
- [53] E. N. Gilbert. “Random Graphs”. In: *The Annals of Mathematical Statistics* 30.4 (1959), pp. 1141–1144.
- [54] P. Erdős and A. Rényi. “On random graphs. I”. In: *Publicationes Mathematicae Debrecen* 6 (1959), pp. 290–297.
- [55] P. Erdős and A. Rényi. “On the evolution of random graphs”. In: *Publicationes Mathematicae, Institute of the Hungarian Academy of Sciences* 5 (1960), pp. 17–61.
- [56] P. Erdős and A. Rényi. “On the strength of connectedness of a random graph”. In: *Acta Mathematica Academiae Scientiarum Hungaricae* 8 (1961), pp. 261–267.
- [57] B. Bollobás. *Random Graphs*. 2nd ed. Cambridge: Cambridge University Press, 2001.
- [58] C. Bron and J. Kerbosch. “Algorithm 457: finding all cliques of an undirected graph”. In: *Communications of the ACM* 16.9 (Sept. 1973), pp. 575–577.
- [59] B. Assarf et al. “Computing convex hulls and counting integer points with polymake”. In: *Mathematical Programming Computation* 9.1 (2017), pp. 1–38.
- [60] *OSCAR – Open Source Computer Algebra Research system, Version 1.5.0*. The OSCAR Team, 2025.
- [61] *The Computer Algebra System OSCAR: Algorithms and Examples*. 1st ed. Vol. 32. Algorithms and Computation in Mathematics. Springer, 2025.